

AcoustiLab v2: Headphone Design and Simulation Workbench Specification

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1. Purpose, audience and scope

AcoustiLab v2 is a browser-native workbench for designing the acoustics of over-ear headphones, accurate enough that a prototype built from its output measures close to prediction below about 2 kHz and qualitatively right above. It replaces the v1 demonstrator, whose three-node lumped model could not support the treble, leakage and audition claims made for it.

Audience. Acoustic engineers at headphone companies, serious DIY designers who build and measure hardware, and students of electroacoustics. The tool assumes the user will eventually own a prototype and a measurement rig, and it is built around that loop.

In scope

- Circumaural and supra-aural passive headphones with dynamic, planar magnetic or electrostatic drivers, one or more drivers per cup.
- A complete electro-mechano-acoustic network that includes a standards-based ear simulator, so output is reported at the drum reference point and the ear reference point, not through a fixed overlay.
- Ear-pad leakage, fit variation and wearing robustness; passive isolation; sound bleed to the room.
- Export of the acoustic plant for active noise cancellation and DSP design.
- Fitting the model to measured impedance and frequency response of a real prototype.
- Auralization derived from the model's own transfer function.

Non-goals

- In-ear monitors and true-wireless earbuds in the first release. They follow later as their own variant with insert-specific fixture, fit and canal models; the element library is shared.
- General 3-D finite-element or boundary-element solving in the browser. The ceiling is 2-D axisymmetric; beyond that a purpose-built open-source companion solver on the same machine or the local network takes over, with no manual export step and no

commercial package.

- Perceptual claims beyond published, peer-reviewed preference models.
- Industrial design, structural mechanics, thermal, battery and radio.

What changed from v1 and why

v1	v2	Reason
Three hand-derived nodes solved by Cramer's rule	Netlist of arbitrary size solved by modified nodal analysis	Elements can be added without re-deriving determinants; schematic and solver cannot diverge; controlled-source sign errors are caught by tests
Fixed ear-canal gain overlay	IEC 60318-4 style ear simulator inside the network, plus alternatives	Canal resonance and drum load interact with the cup; the overlay hid that
Cushion leak as a pure resistance	Perimeter slit with resistance and inertance from viscothermal theory, plus an explicit eyeglass channel	Produces the correct 6 and 12 dB per octave regimes; v1 claimed a slope its model could not make
Lumped only, plotted to 20 kHz without caveat	Fidelity ladder with validity shading, transmission-line segments, cavity modal expansion, diaphragm breakup modes	Treble behaviour comes from physics the model contains, not from copy
Fleece as a fixed 12 % volume increase	Porous fill model scaled by fill fraction, bounded by the isothermal limit	Matches the slider it is attached to
Undefined drive level	Explicit drive convention with sensitivity readouts	Absolute SPL and impedance switching were otherwise meaningless
Audition through five hand-tuned biquads	Convolution with the model's own complex response, level matched	The audition is the physics, not a parallel guess

v1	v2	Reason
Presets named after commercial products	"-like" presets shipped with validation data	Names without validation are decoration
No validation	Analytical, property and measured-reference test suite	A solver that has not passed the pressure-chamber and leak-slope checks is not trustworthy
No robustness, isolation, ANC or fitting	All four present	These are what a working engineer spends most of their time on

Design principles

1. One netlist is the truth. The schematic, the solver, the cross-section geometry and the explain panel all derive from it.
2. Every plot shows where the model is trustworthy, by frequency band and by fidelity level.
3. Every statement the interface makes about cause and effect is computed from the solver, never hard-coded.
4. Measurements in, measurements out. The tool ingests real data, fits to it, and exports what other tools need.
5. Cheap physics first. Closed-form and one-dimensional models wherever they suffice, 2-D axisymmetric finite elements as the in-browser ceiling, and the purpose-built companion solver beyond that.

Product-class scope matrix. An engineer designs one class at a time, and the classes differ in front-chamber topology, leak mechanism, ear load and force convention, so the tool never presents their models as interchangeable.

Class	Front chamber	Dominant leak	Ear load and fixture	Force convention	v2 status
Circumaural	Pinna-enclosing cavity	Perimeter slit plus porous pad	IEC 60318-1 for the Harman fixture class; 60318-4 based heads; P.57 Type 4.3	Human clamp force from the headband model; fixture application force reported separately	Core

Class	Front chamber	Dominant leak	Ear load and fixture	Force convention	v2 status
Supra-aural	Pinna-compressing pad, leak-dominated	Slit under the pad, P.57 Type 3.2 grades	IEC 60318-1; Type 3.3	As above	Supported variant
Insert and true wireless	Occluded canal residual volume, nozzle and tip	Tip seal; ETSI Annex C port	60318-4 with extender; Type 4.3 required for deep insertion; 60318-8	Insertion depth	Planned after supra-aural and open wearable; shared element library
Open wearable stereo: open-ear buds and clip-ons	No enclosed chamber; a near-field source beside the pinna driving the open canal	No seal; bass set by standoff and dipole balance; bleed is a design variable	Open-ear load: canal plus radiation impedance on a head; Type 4.3 or 4.4 head in free field	Wearing position and standoff distribution	Second variant after supra-aural, prioritised over in-ear; needs the companion solver's exterior boundary elements for bleed and natural pinna incidence

Personas. The acoustic engineer who tunes a vent to a bass target under excursion and isolation constraints and needs validity bands, fit statistics and measurement fitting. The DSP or ANC engineer who exports the plant with true phase and an uncertainty set. The quality engineer who exports production limits. The DIY designer or student who learns from presets, the explain panel and audition. Section 19 states what each can do at each release and how it is tested.

2. Modeling philosophy: the fidelity ladder

The tool runs one netlist at five selectable fidelity levels, L0 to L4, computes the validity of every element from its geometry with one normative criterion table, and draws two bands on every curve: the level's model-form band and the Monte Carlo parameter band. A lumped element is credible while its largest dimension is a small fraction of a wavelength; for a 60 mm front cavity the error reaches 3 percent near 270 Hz, 10 percent near 455 Hz and 3 dB near 910 Hz, and by 3 kHz the cavity has its first standing wave. v1

ignored this and described treble effects it could not compute. v2 makes the ceiling visible, raises it where physics allows, and replaces the least-trusted part of the model with measurement when it does not.

Level	What is distributed or measured	What it newly captures	Credible ceiling, typical over-ear	Cost per sweep
L0 Lumped	Nothing; every element lumped but thermoviscous-corrected: leaks and vents as resistance plus inertance, cavities with wall loss, fill as lossy compliance	Pressure-chamber bass, leaks, vents, Helmholtz resonances, impedance, sensitivity, isolation plateau, excursion, low-frequency Monte Carlo	Cavity criterion at 10 percent error, about 450 Hz for 60 mm; piston ka of 1 near 2.7 kHz	Under 1 ms

Level	What is distributed or measured	What it newly captures	Credible ceiling, typical over-ear	Cost per sweep
L1 Distributed closed-form, the default working tier	Ducts, slits and annuli as thermoviscous two-ports; canal and ear simulator as transfer-matrix chains; radial front-chamber line; rigid-wall modal sums for cup and front cavity with wall admittance and position-dependent driver coupling; porous equivalent fluids; mesh and perforate impedances; radiation loads; driver piston plus surround branch or imported modes	Canal quarter-wave, pipe and depth modes, transverse cavity modes, driver position, breakup peaks	About 8 kHz in shape for idealised geometry; above about 5 kHz bands only, because pinna diffraction is not modelled	Milliseconds
L2 Reduced-order numerical, optional and after the first release	2-D axisymmetric finite elements with boundary-layer wall conditions for idealised cups, reduced to an N-port matrix	Non-separable cup geometry, liner placement	8 to 10 kHz for near-axisymmetric cups; the pinna breaks symmetry above about 4 kHz	0.1 to 5 s; under 100 ms after eigen-reduction

Level	What is distributed or measured	What it newly captures	Credible ceiling, typical over-ear	Cost per sweep
L3 Companion solver	N-port matrices computed by the purpose-built companion solver of Section 9, an open-source finite-element and boundary-element service on the same machine or a workstation on the local network, which receives geometry and boundary data from the browser automatically and returns results checked for reciprocity and passivity	Everything three-dimensional	Three-dimensional cups with pinna to about 10 kHz, set by mesh and memory; bands above about 5 kHz	Seconds to minutes on the workstation, asynchronous and cached

Level	What is distributed or measured	What it newly captures	Credible ceiling, typical over-ear	Cost per sweep
L4 Measured hybrid	Imported measured responses and impedances of a driver on a baffle, a coupler, a prototype cup or an ANC plant, used as elements or as replacements above the model's valid frequency, with fitting of the identifiable parameters	The real device above 2 kHz	The measurement's own bandwidth and uncertainty	Negligible

The ear load has its own ladder: a lumped 60318-4 chain valid to about 3 kHz, the transfer-matrix chain with damped variant valid to 10 kHz as a human ear and 16 kHz as a coupler, the Type 4.3 canal with fitted drum from 20 Hz to 20 kHz with published spread, and population variation only with licensed data. The driver ladder D0 to D4 is in Section 5.

Normative validity criteria. One table drives every shading, badge and advisory limit. Numbers use 343 m/s.

Element or assumption	Criterion	Levels and examples
Cavity as one node	kL , with L the largest distance from the driver centroid to the observation point	kL of 0.3, 0.5 and 1.0 give 3, 10 and 36 percent error, or 0.3, 0.9 and 3 dB; for 60 mm that is 270, 455 and 910 Hz
Duct or canal as a plane-wave line	First transverse cut-on at $1.8412 c$ over $2\pi a$	26.8 kHz for a 7.5 mm canal; 3.35 kHz for a 60 mm cavity treated as a duct

Element or assumption	Criterion	Levels and examples
Slit or tube loss regime	Shear wavenumber, transverse size over viscous layer	Below 2 Poiseuille; 2 to 10 full thermoviscous; above 10 resistance grows as the square root of frequency
Thermoviscous line model	Stinson's bound, radius times frequency to the 1.5 under 1e6 in cm and s	About 19 kHz for a 7.5 mm canal, 16 kHz for a 10 mm duct, 46 kHz for a 2 mm hole; flagged, not silently exceeded
Rigid piston	ka of 1	About 3 kHz for a 10 cm ² diaphragm
Leak path as lumped	Path length under a tenth of a wavelength	A 25 mm pad face gives 1.4 kHz
Driver breakup	Imported first mode, or the ring-frequency label as order of magnitude	Film domes typically 3 to 6 kHz, unconfirmed; treated as a user input
Ear simulator as a human ear	Scope of the standard	60318-4 human-valid 100 Hz to 10 kHz, coupler-extrapolated below and coupler-only above; Type 4.3 20 Hz to 20 kHz
Finite-element mesh	Element size under a sixth of the wavelength for quadratic elements and the pollution bound on wavenumber cubed times size squared	Refined to 0.5 mm at ports and leaks

Rules the implementation must follow

1. Shading is computed, never assumed. It begins at the lowest 10 percent frequency among the active elements in the signal path and deepens at the 36 percent frequency; ear-simulator scope and driver breakup add their own bands.
2. Switching levels never changes the netlist. A cavity is one object whose representation the level chooses, and a distributed representation replaces its lumped node rather than sitting in parallel with it.
3. L0 is always live. Interaction recomputes L0 immediately and draws the previous higher-level result as a ghost until its recompute lands.

- 4. Above the ceiling the tool plots but never advises. Readouts, sensitivities, optimiser objectives and the explain panel restrict themselves to the credible band unless the user widens it deliberately.
- 5. Every curve carries both bands, and a continuity test requires that switching levels changes results below the lower level's 3 percent frequency by under 0.1 dB.
- 6. The internal range is 10 Hz to 40 kHz with 20 Hz to 20 kHz as the default view; bass is where the model is most accurate and where leaks and vents show.

Why not finite elements everywhere. A three-dimensional thermoviscous model of a cup with a pinna is tens of thousands to a million degrees of freedom per frequency and needed roughly 30 to 100 GB of memory to converge above 5 kHz in a published vendor example. Design happens at L0 and L1 because they answer in milliseconds, L2 confirms a candidate, L3 handles the final geometry, and L4 is how practitioners actually get above 2 kHz. The ladder mirrors how headphone engineers work.

3. Solver architecture

The solver is a generic modified nodal analysis over a typed multi-domain netlist, assembled and factorised per frequency in a WebAssembly core. Nothing about the headphone is hard-coded in the solver; the netlist carries it all, and the schematic, cross-section and explain panel are views of the same netlist.

Analogy and domains. The impedance analogy is used in all three domains: electrical (voltage, current), mechanical (force, velocity) and acoustic (pressure, volume velocity). Mass and inertance are inductors, compliance is a capacitor, loss is a resistor. Each domain has its own ground: circuit return, fixed frame, ambient pressure. The moving-coil coupling is a gyrator with parameter Bl , so force equals Bl times current and back-EMF equals Bl times velocity. The diaphragm coupling is an ideal transformer with ratio S_d , so volume velocity equals S_d times velocity and force equals S_d times pressure difference. Planar and electrostatic drivers use the same two couplers with different parameter sources.

Element kinds

Kind	Stamped as	Examples
Two-terminal impedance $Z(\omega)$	Admittance between two nodes	R, L, C, viscothermal tube, radiation load, lossy inductor, mesh, porous layer

Kind	Stamped as	Examples
Two-port	ABCD transfer matrix converted to a 2×2 admittance block; falls back to an impedance-form stamp when the conversion is singular	Transmission-line segment, ear simulator, imported measured two-port, FEM-derived block
Controlled sources and couplers	Extra branch variables in the MNA border	Gyrator (BI), transformer (Sd), voltage source with series impedance
Macro	Expands into a sub-netlist before assembly	Dynamic driver, planar driver, IEC 60318-4 simulator, pad with leak and porosity, vented cup
Probe	No stamp; names an output	Pressure at drum, voltage at coil, volume velocity in a branch, diaphragm velocity

Assembly and solve. For N nodes and M extra branch variables the system is $(N+M)$ square, complex, dense, typically 20 to 80 unknowns. Dense partial-pivot LU in double precision beats sparse methods at this size and is deterministic. Structure and stamping order are fixed once per netlist; only values change with frequency, so the per-frequency cost is the numeric fill plus factorisation. Sweeps are split across a Web Worker pool by frequency.

Conditioning. Electrical, mechanical and acoustic magnitudes differ by up to ten decades, so the matrix is row and column equilibrated before factorisation and the condition number is monitored; without this, pivoting loses four to six digits near resonances, and a hand-derived Cramer's rule solution is neither general nor backward stable. Two-ports are always lossy: a lossless tube's transfer matrix has a zero off-diagonal term at every half-wave frequency and cannot be converted to admittance form there. Frequency sampling is seeded at the poles of the rational fit described below so no high-Q feature falls between grid points.

Units and conventions sheet. Normative for every element, import and export, because every sign and unit ambiguity found while preparing this specification, gyrator phase flips, a sixteen-fold error from a wrong hydraulic radius, the $6/5$ slit-mass factor, micrograms for milligrams, rayl for rayl per metre, is a silent wrong-answer generator.

- Time convention e to the plus jw everywhere; the thermoviscous square roots, the boundary-layer wall condition and the modal damping sign all follow from it.
- Positive volume velocity flows into a node; two-ports use the transfer-matrix convention with the output flow leaving the port.
- SI internally with unit-suffixed keys; display units are MKS rayl in $\text{Pa}\cdot\text{s}/\text{m}$ for meshes and fabrics, $\text{Pa}\cdot\text{s}/\text{m}^3$ for acoustic impedance, $\text{Pa}\cdot\text{s}/\text{m}^2$ for flow resistivity, and a CGS rayl is ten MKS rayl. The porous-model variable is defined once as frequency over flow resistivity and coefficients are converted to it.
- Reference points per ITU-T P.57 and P.58; levels in dB SPL re $20\ \mu\text{Pa}$ at the drum reference point; sensitivity per IEC 60268-7 at 500 Hz in dB per volt, with dB per milliwatt equal to dB per volt minus $10\log_{10}$ of 1000 over rated impedance; curves normalised at 500 Hz; left and right named, never averaged silently; phase lag negative; fractional-octave bands per IEC 61260-1:2014; the feedback sensitivity written as one over one plus loop gain.
- Sensitivities are reported in decibels per percent, the per-unit derivative divided by 100.

Frequency grid policy. One adaptive log grid is the solver grid: base 512 points from 10 Hz to 40 kHz, bisection refinement to 0.1 dB and 2 degrees, seeded at the poles of the rational fit. The uniform grid for audition is obtained by re-solving on it, never by interpolation. Impulse-response length follows the decay time of the lowest-frequency high-Q pole, about 2.2 times Q over frequency. The realised grid is exported with every result.

Direct-current handling. Floating acoustic nodes and inductive loops make the matrix singular as frequency tends to zero, so the assembler adds a minimum conductance or excludes zero frequency, and the unwrapped-phase reference is defined at the lowest solved frequency.

Engine as a library. The solver exposes a stable JSON contract and runs headless under Node as a command-line tool for batch sweeps, design-of-experiments runs and continuous-integration regression of designs. The interface is one client of the engine. For the first release sensitivities come from central finite differences reusing the factorisation; the adjoint method follows.

Frequency-dependent elements. Each element implements $Z(\omega)$, $Y(\omega)$ or $ABCD(\omega)$ and, where cheap, analytic derivatives with respect to its parameters. Thermoviscous elements need Bessel functions of complex argument; the core implements them with series and asymptotic forms tested against reference values to $1\text{e-}9$.

Sensitivities. In the full design, output derivatives with respect to every parameter come from the adjoint method, one extra back-substitution per frequency per output, which powers the tornado charts, the explain panel and gradient-based fitting. The first release uses central finite differences that reuse the factorisation, as the engine paragraph below states; the adjoint path replaces them in phase 2 without changing any output.

Fitting and optimisation. Levenberg-Marquardt over log-transformed bounded parameters, with an objective built from weighted dB error on magnitude responses plus impedance magnitude and phase. CMA-ES is available for multimodal problems and for discrete choices such as mesh grades from the database. Tolerance analysis uses Latin hypercube sampling of parameter distributions and reports response envelopes.

Time domain and plant export. Many elements are irrational in frequency, so the linear network is fitted to a rational model by vector fitting (Gustavsen and Semlyen, 1999) of order 20 to 80, which yields a state-space realisation for export and for time-domain simulation. Nonlinear driver elements are then integrated in the time domain against that rational load with a trapezoidal or TR-BDF2 scheme for distortion estimates.

Impulse responses. The network is re-solved on a uniform grid, made Hermitian with real DC and Nyquist bins, and inverse transformed; a minimum-phase variant is produced by the cepstral method under the excess-phase policy of Section 16, and group delay comes from the rational model rather than from wrapped phase.

Sign convention test. Positive source voltage at zero frequency drives positive coil current, positive force toward the ear, positive front-cavity pressure and positive pressure at the drum probe. This single end-to-end check is a unit test, because controlled-source sign errors are the most common bug in hand-built electroacoustic solvers.

Implementation. Rust compiled to WebAssembly for assembly, factorisation, special functions, vector fitting and the optimisers; TypeScript for the netlist model, UI and Web Audio. Results are bit-reproducible across runs, which makes golden-file regression tests possible. The netlist JSON schema is given in the appendix.

4. Electrical domain and drive conventions

Every SPL figure in the tool states its drive: voltage, source impedance and ear load. Without those three an SPL curve is meaningless, which is why IEC 60268-7 defines a characteristic voltage rather than a bare level. v1 had no drive convention, so switching between a 32 Ω and a 300 Ω motor produced an unexplained 10 dB step.

Source. A voltage source with a selectable output impedance: 0 Ω ideal, presets for a sub-ohm portable source and a 10 Ω desktop amplifier, the legacy 120 Ω figure that derives from IEC 61938 and is marked as such, and a custom value. Cable and connector resistance are optional series elements that matter for 16 to 32 Ω loads. The source carries voltage and current clipping limits, a dBFS-to-volts gain for DSP-driven products, balanced or single-ended output, and battery-voltage headroom for active designs, because those products are limited by amplifier swing at the excursion-limited bass boost rather than by the transducer. A constant-current mode exists for diagnostics. Source impedance colours the response by $20 \log_{10}$ of $|Z_{in}|$ over $|Z_{in} \text{ plus } Z_{out}|$; a 120 Ω source on a 32 Ω headphone with a 52 Ω resonance peak adds about 3 dB at the peak, and a 10 Ω source under 1 dB. The classic bass bump of a high-impedance open-back on a high-impedance amplifier is this one resistor.

Drive conventions. Three, always labelled:

1. Characteristic voltage per IEC 60268-7: the voltage that produces 94 dB SPL at 500 Hz at the drum reference point of the selected ear simulator through the specified source impedance. This is the default for distortion and level-dependent readouts.
2. One milliwatt into rated impedance.
3. One volt RMS.

Sensitivity is reported in dB SPL per milliwatt and dB SPL per volt together with the conversion, which adds $10 \log_{10}$ of 1000 over rated impedance, and the reference frequency, 500 Hz per the standard or 1 kHz per most datasheets. The rated-impedance check flags any frequency where impedance magnitude falls below 80 percent of the rated value.

Voice coil. DC resistance with the copper temperature coefficient of 0.00393 per kelvin, or 0.00377 for aluminium and copper-clad aluminium wire. Inductance defaults to the datasheet value, which is near zero for many headphone drivers, and is modelled as the two-branch LR-2 network so it is realisable in the time domain; LR-3 extends the band. The Leach power law and the Thorborg-Unruh semi-inductor are offered as frequency-domain impedance fits only, since they are not realisable as lumped networks. For most headphone drivers the inductive corner lies far above the audio band, for example 32 Ω and 50 μH give about 100 kHz, so inductance matters for impedance phase, source interaction and eddy-current damping rather than for SPL, and an invented inductance of a few hundred microhenries would falsely roll off the treble of a 32 Ω driver. Displacement- and current-dependent inductance belong to the nonlinear model in Section 5 and are enabled only with measured data.

Electrical damping and scaling. Electrical Q equals R_e times $2\pi f_s$ times M_{ms} over Bl squared. Rewinding a coil at constant mass scales Bl with the square root of R_e , so Bl squared over R_e and therefore the response shape are invariant; only level moves. Going from 32 to 300 Ω costs 9.72 dB at constant voltage, gains 9.72 dB at constant current and changes nothing at constant power. The tool shows this as a live readout when coil resistance changes so the level step is explained, not mysterious.

Power and thermal. Coil current, dissipated power and diaphragm excursion are computed for the active drive. A first-order thermal model with a thermal resistance and time constant raises R_e with dissipated power, so long-term power compression and the drift of resonance damping are visible. Coil power above a user-entered rating and excursion above the driver's linear limit are flagged.

Advisory limit check. For the selected programme-simulation noise, the tool reports the A-weighted level at the drum at a given source voltage against the EN 50332 limits, marked advisory. The full regulatory panel is specified in Section 10.

Other transducer types. Planar magnetic drivers present a nearly resistive, nearly flat impedance with microhenry-scale inductance. Electrostatic drivers are a capacitance behind a step-up transformer with a bias supply and are driven by voltage swing, not power. Both reuse the same source and convention machinery; their coupling models are in Section 5.

5. Transducer models

The driver is a rigid piston plus a small set of modal branches, never a single mass, and every driver record carries provenance and the loading condition under which its parameters were measured. Moving-coil drivers are first-class. Planar magnetic and electrostatic drivers ship as experimental plug-ins behind the same mechanical and acoustic interface, because no peer-reviewed lumped model for planar headphone drivers was found and the electrostatic derivation needs checking against a measured unit before it is trusted.

Driver sub-ladder

Level	Model	What it adds
D0	Thiele-Small piston: R_e , Bl , M_{ms} , C_{ms} , R_{ms} , S_d	Bass, coupled resonance, impedance peak

Level	Model	What it adds
D1	Plus lossy inductance and source impedance	Impedance phase, electrical damping, amplifier interaction
D2	Plus two-degree-of-freedom dome and surround branch	The surround anti-resonance dip and edge peak in the 1 to 6 kHz region; complex effective area
D3	Plus imported modal data or a measured velocity-weighting filter	Breakup peaks and their coupling to cavity modes above about 3 kHz
D4	Plus nonlinear and thermal parameters in the time domain	Distortion trends versus level, DC offset, excursion limits, power compression

Piston branch. Standard small-signal equations: force equals Bl times current, back-EMF equals Bl times velocity, and the mechanical impedance is the sum of mass, loss, compliance and the front and rear acoustic loads reflected through S_d squared. Moving mass, loss and compliance include the air load and differ between free air, coupler and vacuum, so the record stores which applies. Effective area is not the geometric dome area: for one-piece film diaphragms the rim moves and radiates, so S_d should come from a volume-displacement or laser measurement. The piston range ends near ka of 1, about 3 kHz for a 10 cm^2 driver, independent of stiffness.

Reference driver. The Tympany HPD-40N16PET00-32 is the only 40 mm headphone driver found with a complete public datasheet and is the datasheet-anchored reference for tests and the default preset: R_e 32.8Ω within 5 percent, minimum impedance 33.5Ω , inductance listed as zero, M_{ms} 0.3 g, f_s 81.8 Hz within 15 percent, Q_{ms} 2.71, Q_{es} 1.01, Q_{ts} 0.74, Bl 2.42 T·m, S_d 10 cm^2 , X_{max} 0.8 mm, rated noise power 10 mW. Its compliance is printed as $11.7 \mu\text{m}/\text{N}$ where mm/N is the only value consistent with the resonance, which is why every imported parameter set is cross-checked against the resonance identity before use.

Typical ranges for sliders. Free-air resonance 50 Hz to 1 kHz for headphone transducers, moving mass 0.1 to 0.5 g, force factor 0.3 to 2.5 T·m, coil resistance 16 to 300Ω . Headphone drivers often have small Bl squared over R_e and low mechanical Q ; when the ratio of mechanical to electrical Q falls below about 0.1 the impedance hump is too small to identify parameters electrically, and the tool says so.

Inductance. Default is the datasheet value, which is near zero for many headphone drivers, with the two-branch LR-2 network for anything that must run in the time domain. LR-3 is available for wide-band fits. The Leach power-law and the Thorborg-Unruh semi-

inductor are frequency-domain fits only, since they are not realisable as lumped networks. Displacement- and current-dependent inductance are enabled only with measured data, because for micro drivers they are usually negligible. The impedance plot shows the ideal 6 dB per octave inductor line so the real 3 dB per octave rise is visible.

Creep. Film suspensions lose up to half their stiffness at very low frequency relative to the value at resonance. Compliance follows a log-frequency law with a creep factor defaulting to zero, after Knudsen and Jensen 1993, and is exposed in the impedance fit so measured impedance below resonance can be matched.

Dome and surround. A two-mass model couples the dome-plus-coil mass to an effective surround mass through the surround's bending stiffness, with the surround also sprung to the frame. Volume velocity is the sum of two areas times two velocities, so the effective area is complex and dips where the surround moves in antiphase. This is a lumped approximation of the first non-piston mode and is standard mechanics rather than a headphone-specific published model.

Breakup. Each mode is a resonant branch with modal mass, damping ratio and an effective modal area that may be small, zero or negative, feeding its own volume velocity into the front and rear nodes. Modal parameters come from imported scanning-vibrometer or finite-element data, or from a simple clamped-dome shell model run in the same WebAssembly solver. The ring-frequency estimate from material properties is displayed only as an order-of-magnitude label, because thin film domes break up well below it. A frequency-dependent fudge on effective area is not accepted as a substitute, since it ignores how the mode couples spatially to the cavity. Rocking modes are non-radiating but appear in impedance and cause coil rub; an optional diagnostic takes user-entered asymmetry percentages in stiffness, mass and field and reports the voltage at which the coil touches the gap, since single-suspension headphone drivers offer no reliable mechanical protection.

Driver-level acoustics. The record includes the driver's own rear openings, pole vent and damping paper as resistance-plus-inertance elements, and any front dust screen, so the driver's Q in free air is reproduced before the cup network is added.

Nonlinear and thermal, D4. Polynomials to eighth order in displacement for force factor, stiffness and inductance, plus current-dependent inductance, velocity-dependent mechanical loss, and displacement-dependent area, with the reluctance force of half the current squared times the inductance gradient. Each polynomial carries the excursion range over which it was identified and is clamped beyond it. Asymmetric stiffness, the dominant nonlinearity of rim-suspended film domes, produces DC offset, which is reported. Excursion limits follow the 82, 75, 10 and 10 percent thresholds on force factor,

compliance, inductance and Doppler used by IEC 62458 and Klippel practice, with the 10 percent distortion criterion of AES2 as an alternative, and a separate coil-to-gap contact limit. Thermal behaviour uses a two or three time-constant network with the copper coefficient of 0.00393 per kelvin, so resistance doubles after a 254 K rise; the coil time constant is under a second, convection is off by default, and results above the rated power are labelled as abuse predictions. Distortion is reported as trends against level, not as absolute percentages, and the interface states that excursion at 100 dB in a sealed cup is a few micrometres.

Multi-driver and hybrid designs. A cup may hold several drivers with independent electrical ports, a passive crossover library of resistors, inductors, capacitors and autoformers, or a DSP crossover of biquads or FIR filters with stated latency. Each driver has its own acoustic path into the front cavity, so coaxial tweeters, offset woofers and planar-plus-dynamic hybrids sum at the drum with true phase, and polarity and delay alignment are parameters.

Planar magnetic plug-in. Force factor is the tangential field component along the trace times total active trace length; coil resistance follows from trace geometry; inductance is microhenries so the impedance is nearly resistive and voltage drive equals current drive. Moving mass is film plus conductor plus a large air load. Stiffness comes from tension or, for zero-tension designs, mostly from the air load; tensioned membranes carry Bessel-root modes that appear as notches. The magnet array's slots load each face as resistance plus inertance. The distributed force excites modes through field non-uniformity. Public anchors are a 20 Ω , 103 dB per milliwatt commercial planar and a 32 Ω one; the model is validated against a measured unit before leaving experimental status.

Electrostatic plug-in. Push-pull constant-charge operation linearises the force to the bias times signal voltage times permittivity and area over gap squared. The electrical port sees half the single-gap capacitance plus cable capacitance behind a step-up transformer and a megohm bias resistor. Stability requires the tension stiffness to exceed twice the negative electrostatic stiffness, and linearity requires displacement under a tenth of the gap. A public anchor is a 110 pF, 580 V bias headphone rated 101 dB at 100 V RMS. Balanced-armature receivers are out of scope for the circumaural core.

Diaphragm material library. Modulus, density, Poisson ratio, thickness, loss factor and glass transition for PET, PEN, PEEK, PEI, aluminium, titanium, magnesium, beryllium and bio-cellulose, with unknown fields left visibly blank. The tool computes areal mass and the ring-frequency label; measured modal loss factors for films were not found and must be imported.

Record governance. Every driver record stores source, date, licence and measurement condition. Automatic checks verify the resonance identity within 5 percent, electrical Q from B_l , R_e and M_{ms} , and equivalent volume from area and compliance, and flag unit anomalies such as micrograms where milligrams are meant, a mistake found in a published parameter set during this work. User-contributed records are marked as such.

6. Acoustic element library

Every acoustic element is a complex, frequency-dependent one-port or two-port computed from geometry and air state through the thermoviscous functions. A fixed resistor or inductor exists only as an explicit user override, and the familiar closed-form limits serve as unit-test oracles rather than as elements. At headphone scale the viscous boundary layer is 0.49 mm at 20 Hz, 0.22 mm at 100 Hz, 0.069 mm at 1 kHz and 0.022 mm at 10 kHz, the thermal layer is 1.19 times thicker, so every gap, pore and slit in a headphone is within a few boundary layers somewhere in the band.

Canonical representation. Tubes, slits and other ducts use the Zwikker-Kosten solution as reduced by Stinson 1991: a complex density from the viscous function and a complex compressibility from the thermal function, Bessel form for circles, hyperbolic-tangent form for slits, shape factors for other sections. The shear wavenumber, the transverse dimension over the boundary layer, selects the regime automatically: below about 2 the element is Poiseuille resistance plus a mass with the $4/3$ or $6/5$ factor, above about 10 the resistance grows with the square root of frequency and the mass is geometric.

Christensen's 2024 continued-fraction ladders, passive networks of positive constant elements, are the stored form, so the same element runs in the frequency-domain solve and in a time-domain filter without Bessel functions. Special functions switch to asymptotic forms above a shear wavenumber of about 30 with a continuity test across the switch. A negative real part means the wrong square-root branch and fails the passivity test.

Element	Model	Inputs	What the interface shows
Unfilled cavity	Complex compliance with wall-layer correction, real part up by ϵ and loss tangent ϵ , where ϵ equals $(\gamma - 1)$ times thermal layer times wall area over twice the volume	Volume from geometry; wall area from geometry or a surface factor of 1 to 5 for ribbed cups	Cavity Q of 1 over ϵ , typically 30 to 200, so wall loss alone never flattens a cavity mode
Filled cavity	Leach 1989 network: adiabatic compliance in parallel with a series thermal branch whose compliance is 0.4 times the adiabatic value times fill fraction, corner from fibre spacing after Tarnow 1996 and Lafarge 1997, plus flow resistance in series with the fill air mass	Fill fraction, fibre diameter, bulk density or flow resistivity	Compliance gain versus frequency, bounded by plus 40 percent of the filled air and falling to zero above the thermal relaxation frequency, with its loss
Tube and port	Circular thermoviscous two-port	Radius, length, wall condition	Shear wavenumber and regime; fully lossy at all times
Slit and thin layer	Low-reduced-frequency slit two-port after Tijdeman 1975 and Beltman 1999	Gap, breadth, depth	Regime; for gaps under one boundary layer the squeeze-film limit

Element	Model	Inputs	What the interface shows
Diaphragm-to-grille gap	Radial transmission line of the slit type with the grille perforation as a distributed shunt after Homentcovschi and Miles 2004	Gap, grille open area, hole diameter	Mechanical damping in N·s/m next to the driver's own Rms; a 0.5 mm gap under a 40 mm diaphragm adds about 0.1 N·s/m, comparable to Rms, while a 2 mm gap adds almost nothing
Mesh and screen	Datasheet specific flow resistance over covered area, plus a pore-level tube model calibrated to that value for mass and square-root-of-frequency rise, plus the Ingard-Ising velocity-dependent resistance	Grade from the database or user rayl value, area, thickness, open area	Resistive-flat or frequency-dependent badge: meshes of 30 rayl and above are flat across the band; coarse meshes under 10 rayl are not. A warning when pore velocity exceeds about 1 m/s
Membrane vent	High-resistance layer plus membrane mass and compliance	Datasheet rayl and thickness	Insertion loss against a back cavity; the resonance explains why loss is not monotonic in rayl
Perforated plate	Maa 1998 with Crandall per-hole form and Ingard 1953 end corrections reduced by neighbour interaction	Hole diameter, thickness, open-area ratio	Perforate constant and regime: resistive below 1, transitional to 10, mass-dominated above

Element	Model	Inputs	What the interface shows
Helmholtz vent	Tube two-port with per-end corrections, viscous, radiation and mesh losses, into the cavity compliance	Hole count, diameter, wall thickness, flange condition, mesh	Resonance, Q, and whether the vent is resonant or resistive at resonance
Aperiodic vent	Vent whose mesh resistance exceeds its mass reactance across the band	As above	First-order corner with the cavity
Radiation load	Exact piston in an infinite baffle with Bessel and Struve functions; free-space variant with half the resistance and the $0.61a$ end correction	Aperture, flange condition	A finite-baffle warning of about 3 dB ripple where cup dimension approaches half a wavelength, 1 to 3 kHz
Area discontinuity	Karal 1953 inertance at every tube-to-cavity step	Area ratio	Applied at every two-port junction
Porous layer	Johnson-Champoux-Allard, or its Lafarge extension, as the primary equivalent fluid; Delany-Bazley and Miki as one-parameter fallbacks	Porosity, flow resistivity, tortuosity, viscous and thermal lengths, thermal permeability; or flow resistivity alone	Surface impedance of a rigid-backed layer; an out-of-range badge outside the fallback's validity window
Wall admittance for the modal and finite-element levels	Berggren, Bernland and Noreland 2018 boundary-layer condition	Air state	Sign fixed by the global time convention; a test asserts that positive admittance lowers pressure at resonance

Element	Model	Inputs	What the interface shows
Cup shell and baffle	Surface mass law with panel compliance	Wall thickness, material	Structural resonance region flagged, typically 1 to 5 kHz
Radial front-chamber line	Bessel radial waveguide between diaphragm rim and pad wall	Radii, depth	The 1.5 to 3 kHz front-chamber standing wave the lumped node cannot show
Tabulated and expression elements	User expressions, tabulated impedance or transfer matrices with interpolation and passivity check; imported measured responses	Files	The measured-hybrid level L4 of Section 2

End corrections. Flanged end $0.85a$ from the low-frequency piston limit, $0.8216a$ exact after Nomura et al. 1960; unflanged $0.6133a$ after Levine and Schwinger 1948; array reduction after Ingard; inclined exits after Su et al. 2022. Omitting corrections on a 2 mm hole in a 1.5 mm wall doubles the mass and lowers the resonance by 30 percent. A worked vent: 2 mm hole, 1.5 mm wall, 60 cm^3 cup tunes to 230 Hz with Q near 13 unmeshed; a 160 rayl mesh raises the resistance thirty-fold above the mass reactance and turns the resonator into a resistive leak. Mesh choice, not geometry, decides whether a vent is resonant.

Porous-model conventions. The Delany-Bazley variable is written with air density, frequency over flow resistivity; Miki's coefficients are for frequency over flow resistivity without density. The tool defines one variable and converts coefficients once. The fallback is valid for that variable between 0.01 and 1, which for a flow resistivity of $20 \text{ kPa}\cdot\text{s}/\text{m}^2$ means 200 Hz to 20 kHz; below that Miki or the full model is used. Compression raises pad-foam flow resistivity by about 40 percent at 20 percent strain and is coupled to clamp force. Layered fibres are about twice as resistive through-thickness as in-plane, after Tarnow 2002. Poroelastic frame behaviour is not modelled; a rigid-frame equivalent fluid with compression-dependent flow resistivity is sufficient and identifiable for consumer foams.

Datasheet traps. The air-permeability column on mesh datasheets is measured at a pressure that puts the pores deep in the nonlinear regime and yields resistances an order of magnitude too high; only the rayl value is used. Specific flow resistance in $\text{Pa}\cdot\text{s}/\text{m}$, acoustic ohms in $\text{Pa}\cdot\text{s}/\text{m}^3$ and flow resistivity in $\text{Pa}\cdot\text{s}/\text{m}^2$ are three different quantities, and a CGS rayl is ten MKS rayl; the units panel enforces the distinction.

Air state. Temperature 20 to 37 °C and static pressure are inputs; density, viscosity, conductivity and sound speed follow, and every element's boundary layers are recomputed. Speed of sound rises 0.17 percent per kelvin, density falls 0.34 percent per kelvin at fixed pressure, adiabatic compliance is independent of temperature, and altitude scales every compliance through static pressure. A diagnostics panel lists the two boundary-layer thicknesses at 100 Hz, 1 kHz and 10 kHz beside each element's characteristic dimension.

Element tests. Tube impedance against the exact solution for shear wavenumbers from 0.1 to 100; cavity compliance against the isothermal and adiabatic limits; Helmholtz frequency and Q against Ingard's 1953 measured cases; mesh resistance against datasheet values; perforated-panel absorption against Maa's 1998 figures; porous-layer impedance against the Miki closed form; end corrections at the zero-frequency limit.

7. Ear, head and fixture models

The ear is a network element, not an overlay. The headphone's front-cavity node drives a canal two-port terminated by a drum impedance, and the solver reports pressure at the drum reference point, at the open canal entrance and at the blocked entrance separately. The commonly quoted "ear canal gain" is the ratio of two of these and depends on the headphone's own source impedance, which is exactly why v1's fixed curve was wrong.

Selectable ear loads

Load	Construction	Validity	Use it for
IEC 60318-4 occluded-ear simulator	Three lossy tube sections in cascade with two shunt Helmholtz branches of slit resistance and inertance and side-cavity compliance, terminated by the microphone's equivalent volume. Topology after Nielsen et al. 2004, dimensions from the micro-CT study of Luan et al. 2019, element values fitted to the standard's Table 1 rather than trusted from slit heights whose cube is uncertain	Human-valid 100 Hz to 10 kHz; 20 to 100 Hz coupler-extrapolated because the standard does not validate it there; 10 to 16 kHz coupler-only; the undamped 13.5 kHz half-wave resonance leaves about 10 dB of uncertainty above 10 kHz	GRAS 43AG, 45CA and KEMAR fixtures with an RA0045 class coupler, and the in-ear Harman target
Damped high-frequency 60318-4	Same geometry with damping tuned so the 13.5 kHz peak drops about 14 dB while the 10 kHz level moves under 0.3 dB	As above plus a 2.2 dB tolerance band to 20 kHz; quarter-inch variants are compatible with the standard, not compliant	GRAS RA0401 and RA0402 class couplers

Load	Construction	Validity	Use it for
ITU-T P.57 Type 4.3	Conical thermoviscous transfer-matrix canal built from the standard's public geometry in Table 6 and Annex B, an inclined drum plane, and a drum impedance with an exposed middle-ear cavity compliance fitted to the standard's Table 5-c transfer-impedance target: effective volume 1.63 cm^3 and $27.7 \text{ MPa}\cdot\text{s}/\text{m}^3$ at 500 Hz against 1.26 cm^3 and 35.9 for the 60318-4. The published digital-twin model of Nielsen and Herring Jensen 2022 is a literature cross-check only	20 Hz to 20 kHz by the standard; the thermoviscous line model's own bound is about 19 kHz and is flagged	Brüel and Kjær 5128 with the Type 4620 simulator; human-like prediction
ITU-T P.57 Type 4.4	Type 4.3 pinna, concha and canal entry joined to a 60318-4 based inner canal	As its parts	HEAD acoustics fixtures
KEMAR-style Type 3.3	The 60318-4 network plus a 7.5 mm bore cylindrical extension, 10.0 mm as stated for Type 3.1 and to be verified for 3.3, plus optional anthropometric pinna canal	As the 60318-4 network	GRAS 45CA with KB5000 pinnae and KEMAR

Load	Construction	Validity	Use it for
IEC 60318-1	Leak-free simulator for supra-aural and circumaural earphones; the ear simulator of the Harman over-ear fixture according to Olive 2022; impedance of Types 1 and 3.2 tabulated in P.57	20 Hz to 10 kHz	The Harman fixture class and audiometry; implemented only with values taken from the standard
IEC 60318-8:2022	High-frequency coupler for hearing aids and inserts to 16 kHz	Element values not yet obtained	Insert and hearing-aid comparisons
User-defined	Tabulated drum impedance or a series R-L-C fit, with an optional canal area function fitted from a measured entrance impedance after Wulbusch et al. 2023	As supplied	Individual ears and non-standard couplers

Canal model. Twenty to sixty conical segments, each a 2x2 transfer matrix with complex wavenumber and characteristic impedance from the thermoviscous solution, so wall losses are physical rather than a fitted Q. A straight-axis chain deviates up to 20 percent from a curved canal; the curvature correction of Xia et al. 2024 is offered. Exposed parameters are length, mid-canal area, taper and reference-plane offset; cross-section shape is not exposed because GRAS's 2024 study found it irrelevant while length and volume move the response by 1 to 2 dB and insertion depth by 0.5 to 1 dB.

Drum impedance. The default human termination is a tabulated complex impedance with a middle-ear cavity compliance and user scale factors on its resistive, mass and compliance parts, because the 2 to 6 kHz transfer impedance depends on that cavity volume. Its structure follows the two-piston network of Shaw and Stinson and the measured data of Hudde and Engel 1998 and Voss and Allen 1994. A rigid termination and a

60318-4-equivalent termination are available for diagnostics. Above about 10 kHz the field near the inclined drum is three-dimensional, so the drum reference point is defined as the pressure at the drum-simulator microphone plane.

Reference points and coupling. The tool reports pressure at the drum reference point, at the entrance open and blocked, and the pressure division ratio between them, which equals canal input impedance over the sum of canal and headphone source impedances, following Møller et al. 1995 and Hammershøi and Møller 1996. Blocked-entrance measurements have the lowest inter-subject variation, per Møller et al. 1995. A sealed circumaural cup with a large front volume is a near pressure source, so the canal's quarter-wave gain near 3 kHz survives under the headphone and shifts with cup impedance; with a small residual volume the resonance moves toward the half-wave condition. The tool shows that shift rather than assuming it.

Pinna and concha. These lie outside a one-dimensional model. The user enters concha volume as part of the front volume, and three treatments exist for diffraction above about 4 kHz: none, a concha Helmholtz surrogate at the canal port, or a pinna transfer table computed by the companion solver of Section 9. All three are labelled and a pinna-free result is never presented as a head-and-torso prediction. ISO 11904-2 manikin tables convert drum pressure to free-field and diffuse-field related levels, with the caveat that they are for a 60318-4 manikin and differ from 5128 responses. Pinna hardness of 35 plus or minus 6 Shore OO, measured 15 mm forward of the canal opening, is the P.57 requirement that the 5128 meets; individual manufacturer pinna values were not confirmed. Hardness affects the seal, not the canal.

Population mode. The tool ships the Type 4.3 fitted load with published spread bands and does not claim to predict an individual ear; the in-vivo mean curves themselves are not licensed for redistribution. Monte Carlo draws canal length normally with mean 22 mm and standard deviation 1.9 mm from the reference plane, mid-canal area log-normally with median 47 mm² over a range of 17 to 108 mm², and drum impedance scaled by up to 3 dB, following Jönsson et al. 2018 on 32 adults. Age is a covariate: the first-bend area is about 44 mm² in young adults and 69 mm² in the oldest group per Balouch et al. 2023. Measured inter-subject impedance spread is 20 to 30 percent of the mean below 5 kHz and 50 to 90 percent from 5 to 14 kHz, so a single deterministic treble curve misrepresents what a panel measures.

Fixture details that matter

- The microphone termination is modelled explicitly as a WS2P half-inch capsule with under 20 mm³ equivalent volume. A quarter-inch or low-noise capsule legitimately changes the response above 10 kHz, and the tool reproduces that instead of hiding it.

- Application force on a fixture is reported with every result, as P.57 requires: about 10 N for hard earcaps on the Type 3.3 pinna, 0 to 18 N on the 5128 positioner. It is a different quantity from human clamp force and usually exceeds it.
- A flat-plate coupler such as the 43AG and a head-shaped fixture such as the 45CA or 5128 present different cushion seals; the pad model in Section 8 takes the fixture surface as an input. Nominal ear-to-ear spacing is 132 mm per P.58, and the fixture's own mounting leakage must exceed 35 dB.
- P.57 Table 3-b defines leakage simulation through cylindrical holes for handset ears; it is a leak preset alongside the Type 3.2 slits of Section 8.
- Distortion readouts flag test frequencies whose harmonics fall within 1.5 kHz of the coupler's half-wave resonance, because fixture resonances inflate measured distortion.
- Default air conditions follow IEC 60318-4: 23 °C, 101.325 kPa, 50 percent relative humidity. Temperature is adjustable because ear-side air is body-warmed and shifts resonances by about 2 percent.

Fixtures available to this project. A HEAD acoustics head and torso simulator with ears per ITU-T P.57 Type 3.3 or 4.4 depending on configuration, and an Audio Precision APx517B analyser, both in limited booked sessions, plus a planned IEC 60318-4 class clone coupler. Two features follow. A clone-calibration step fits the 60318-4 network to the specific coupler's measured or supplied transfer impedance, because clones deviate from the standard's table. A scripted measurement protocol turns one booked session into the complete validation set: impedance on and off the fixture, the re-seat series, sweeps at the characteristic voltage, and the predictions pre-computed so the comparison happens on the spot.

Build-time self-test. The 60318-4 network must reproduce the standard's Table 1 transfer-impedance levels at all 21 third-octave frequencies within tolerance, give 1260 mm³ effective volume at 500 Hz and place the half-wave resonance at 13.5 kHz plus or minus 1.5 kHz. The Type 4.3 load must reproduce Table 5-c within its tolerances. A build that fails either does not ship.

8. Ear pad, leakage, fit and robustness

The cushion leak is a sub-network of thermoviscous slit segments around the pad perimeter, not a resistor, and fit is a random variable with named states rather than a single number. Real leaks are localised, the bottom of the cushion usually dominates, and

sub-millimetre gaps sit within a few viscous boundary layers, so their resistance and inertance are comparable and both change with frequency. Nothing in this section asserts a bass slope; the slope falls out of the solved network.

Leak sub-network. At least four and by default eight parallel segments, each a series resistance and inertance computed from gap height, breadth along the perimeter, path depth equal to the pad face width and an end correction, using the exact laminar slit solution so the transition from viscous to inertive behaviour is automatic. The interface shows the resistive corner from resistance and front compliance, and the Helmholtz corner from inertance and front compliance, for the current state. A 0.2 mm gap is resistive across the band with a corner near 80 Hz for 30 cm³; a 1 mm gap is inertive above a few tens of hertz and resonates with the same volume near 550 Hz with a Q well below the naive estimate because resistance grows with the square root of frequency. Leak sensitivity depends on the driver's acoustic source impedance, so open-back designs come out less leak-sensitive without any special case, and a diagnostic overlays source impedance, leak impedance and cavity reactance.

Named fit states

State	Definition	Provenance
Sealed	Zero gap	Reference only
Fixture, flat plate	Leak parameters fitted to a flat-plate coupler; over-seals relative to humans	Manufacturer practice
Fixture, soft pinna	35 plus or minus 6 Shore OO pinna per ITU-T P.57, extended soft area	ITU-T P.57 (06/2021), B&K 5128 data
Standardised low and high leak	Slits of 2.8 by 0.26 mm over 84 degrees and 1.9 by 0.50 mm over 240 degrees	ITU-T P.57 Type 3.2
Standardised insert leak	1 mm port covered by meshes of stated flow resistance	ETSI TS 103 640 Annex C
Human nominal, glasses, hair or beard, high leak	Distributions over segment gaps and leaking fraction	Uncalibrated placeholders until fitted to measurements; labelled as such

State	Definition	Provenance
Worn	Pad foam collapse, cover cracking to finite resistance, seam leaks	Ageing preset for tolerance runs

Robustness is reported as the change of any attribute between the nominal and an impaired state, following the acoustic leak robustness definition in ETSI TS 103 640: bass level at 100 Hz, corner frequency, passive insertion loss and secondary-path change.

Eyeglasses. A temple under the pad lifts the cushion and forms two channels of roughly the temple thickness, 2 to 8 mm, along the pad face width. Each is a rectangular duct with aspect-ratio-corrected viscous resistance and inertance, with a shape factor tied to pad stiffness. No public impedance data exist and the ETSI method for eyeglasses is still under study, so this state is labelled an estimate and excluded from acceptance criteria and scores.

Porous pads. Fabric covers are a flow resistance over the exposed area, with presets of about 30, 200, 600 and 2300 rayl from ISO 9053 textile measurements; velour is expected in the 50 to 600 rayl class but was not measured here. Protein leather is impermeable, so its branch is removed and leakage runs through seams and perforations. Foam layers use flow resistivity times thickness over area plus an inertance, or the full porous model of Section 6 when thicker than about 5 mm. Flow resistivity of pad foam rises about 40 percent at 20 percent compression. The rear-cup fill effect is a separate model and is never represented as extra volume.

Pad mechanics and clamping. Human clamp force, a product parameter derived from headband spring rate and the head-breadth distribution around the 152 mm nominal of ITU-T P.58, is kept separate from fixture application force, which P.57 sets near 10 N for hard earcaps and the 5128 positioner spans 0 to 18 N, and which must be reported with every fixture result. Compression is force over in-situ pad stiffness, which Boyer et al. 2015 show must be measured under realistic load and is frequency dependent. Compression sets front volume as cup volume plus opening area times compressed pad height minus pinna and concha volume, and sets the gap distribution as the head-to-pad mismatch profile minus compression times a shape factor. The soft pad wall is also an acoustic compliance shunting the front cavity. Memory foam stiffens with cold and creeps over minutes, so seal, volume and the cup-cushion resonance drift after donning; a settling time constant is a parameter. Contact pressure caps achievable clamp force, and comfort thresholds are unconfirmed in the literature.

Cup and cushion as a mechanical branch. Cup mass on the cushion spring and damping is the path by which ambient sound, footsteps and cable handling reach the front cavity. For a sealed cup the low-frequency isolation plateau is set by cushion stiffness relative to the front-cavity air spring, after Zwislocki 1955; leaks lower that plateau and raise the cup-cushion resonance. Section 10 reuses this branch for isolation and structure-borne noise.

Supra-aural variant. On-ear designs compress the pinna, have a leak-dominated front chamber and couple to the IEC 60318-1 simulator's flat face. The same leak sub-network applies with the P.57 Type 3.2 slit presets and no enclosed pinna volume.

Fit Monte Carlo. Five hundred to two thousand samples over segment gap heights, log-normal with median 0.1 to 0.5 mm, leaking perimeter fraction from a Beta distribution, clamp force normal with 15 percent spread, an eyeglasses Bernoulli, and fabric resistance within 20 percent. Outputs are median and 5 to 95 percent bands, standard deviation versus frequency and worst case. A lab-protocol mode draws five re-seats and flags a standard deviation above 2 dB from 500 Hz to 5 kHz, mirroring IEC 60268-7 practice. Head-shape statistics from ISO 7250 class anthropometry generate the mismatch profile instead of arbitrary gaps. Re-seat spread above about 4 kHz comes from pinna and canal geometry outside the lumped model and is shown only as annotated uncertainty. Reported leak-induced bass deviations across wearers reach 20 dB or more; the underlying distributions are not public, so the tool includes a fitting tool that turns imported re-seat measurements into empirical distributions.

Consequences the model must expose. Excursion rises with leak for a given voltage, so peak displacement is reported per fit state. Electrical impedance reflects the acoustic load, so impedance per fit state is an output, as used in commercial fit detection. A perfectly sealed cup produces static offset and pressure transients on donning or in aircraft, so a minimum equalisation leak is part of every closed design. Unintended leaks at the pad mounting ring, cable entry and microphone ports are modelled as small fixed segments.

Fixture bias. Flat-plate and hard-pinna fixtures under-represent human leakage; Harman needed a modified pinna and the 5128 uses a soft pinna with an extended soft area for the same reason. Results are always tagged with the fit state and fixture assumed.

9. Enclosure and mid/high-frequency modeling

Every cavity in the netlist is one object with four interchangeable representations: a lumped compliance at L0, a closed-form modal expansion or transmission-line depth at L1, a region of the in-browser 2-D axisymmetric finite-element model at L2, or a region of the

companion solver's three-dimensional model at L3. The representation is chosen by the fidelity level, the parameters do not change, and a distributed representation replaces the lumped node rather than sitting in parallel with it, because adding both double-counts the volume.

Enclosure topologies. Closed, vented, aperiodic and open are netlist patterns, not modes of the solver.

Architecture	Netlist pattern	Behaviour the model must show
Closed	Rear compliance only, plus an unavoidable small leak to ambient	Raised coupled resonance, rear-cavity modes above about 3 kHz unless lined
Vented	Rear compliance with a thermoviscous duct to ambient through a radiation impedance, end corrections included	Helmholtz resonance whose frequency depends strongly on end correction; in the worked example a 3 mm hole in a 2 mm wall on 100 cm ³ tunes to 215 Hz with corrections and 325 Hz without
Aperiodic	Vent covered by a resistive mesh from the material database	Broadened, lowered Helmholtz peak; rear load becomes resistive over a band
Open	Rear grille as perforated plate plus cloth into a radiation impedance	Rear pressure near zero; driver sees mostly its own mechanics; bleed to the room becomes an output
Front-to-rear baffle vent	Thermoviscous duct between front and rear nodes	Shunts front pressure into the rear volume, so increasing its area lowers bass and adds damping; it also forms a Helmholtz resonance with the front volume. A unit test asserts the sign of this effect, because v1 described it backwards

Where lumped cavities stop working. The pressure at the far wall of a closed duct exceeds the driven end by $1/\cos(kL)$, so the lumped error is 1 dB at kL of 0.47 and 3 dB at kL of $\pi/4$. For a 60 mm front cavity that is 430 Hz and 715 Hz; with a centred driver the effective distance halves and the limits double. First rigid-wall modes of a 60 by 45 by 20 mm front cavity fall at 2.86, 3.81, 4.76 and 8.58 kHz. A cylindrical rear cup of 25 mm radius and 25 mm depth has its first azimuthal mode at 4.02 kHz, axial at 6.86 kHz and radial at 8.37 kHz. A centred piston excites only the axisymmetric modes; an offset driver, a pinna or an asymmetric leak excites the rest.

Drive-dependent slopes. Above the cavity-loaded resonance a lumped pressure chamber falls at 6 dB per octave under constant diaphragm velocity, 12 dB per octave under current drive, and 12 dB per octave under voltage drive steepening to 18 above the inductive corner. The fall starts at the coupled resonance of Section 4, typically several hundred hertz to about a kilohertz for circumaural front volumes, which is the first-order reason the pressure-chamber picture holds to about 1 kHz and no further.

L1: transmission-line depth. The depth axis of each cavity and every port, slit and tube becomes a two-port transfer matrix with complex wavenumber from the thermoviscous solution. This captures the first depth mode and all pipe modes exactly for plane waves. Lossless tubes are forbidden: their transfer matrix becomes singular at half-wave frequencies when converted to admittance form.

L1: closed-form modal expansion. The cavity pressure is a Green's function sum over rigid-wall eigenmodes of a rectangular or cylindrical box, driven by a piston of given footprint and position. The zeroth term is exactly the lumped compliance, so L0 is the first term of the L1 modal model. Lined walls add damping through their specific admittance, computed from the porous-liner models of Section 6, using the perturbation form from Morse and Ingard; for heavily damped liners the tool falls back to the finite-element path. The piston coupling integral is closed-form for rectangular footprints and numerical for off-axis circles. Truncating at eigenwavenumbers below three times the highest analysed wavenumber keeps a few thousand modes, which costs milliseconds.

L2: 2-D axisymmetric finite elements. Optional, after the first release, with interactivity targets benchmarked before they are promised. The cup, front cavity and canal are meshed as a body of revolution about the driver axis with quadratic triangles. Element size follows two rules: at most one sixth of the shortest wavelength, and the pollution bound of Ihlenburg and Babuska 1995 that keeps wavenumber cubed times element size squared small, which matters at the high kL of a 60 mm cup at 20 kHz; refinement to 0.5 mm at ports and leaks. Boundary conditions are prescribed velocity on the diaphragm, the boundary-layer wall condition on rigid walls, impedance on liners, leaks and the drum, and

equivalent-fluid properties in porous regions. A 60 by 25 mm domain at 1 mm spacing is 6 to 7 thousand degrees of freedom, at 0.5 mm about 25 thousand. Complex sparse factorisation per frequency runs in tens of milliseconds in WebAssembly, so a 240-point sweep is expected to take one to three seconds across eight workers, and after a one-time eigen-solve of a few hundred modes the reduced model re-sweeps in under 100 ms. The model must pass three gates before results are shown: the closed-duct pressure ratio, rigid-cavity eigenfrequencies within 0.5 percent, and a porous-layer surface impedance against its closed form. A refine-by-two control reports the change at the three largest resonances. Oval cups and the pinna break axisymmetry, so results above 3 to 4 kHz carry a wider band.

Hybrid coupling. Distributed regions connect to the lumped network through ports: the driver face, the canal entrance and every vent and leak aperture. Each region is reduced to a complex N-port impedance matrix per frequency, symmetric by reciprocity, and stamped as an admittance block. A port must be small compared with the wavelength for uniform pressure. The driver face is not small above about 1 kHz, so that port carries the diaphragm velocity shape and returns surface-averaged pressure or modal forces, as Section 5 describes. Offset and angled drivers, a standard treble-tuning technique, enter the L1 modal model as position-dependent coupling coefficients; when a geometry is neither box-like nor axisymmetric the interface says so and points to the L3 companion solver.

Lumped-versus-wave diagnostic. One view overlays the lumped, modal and finite-element predictions of drum pressure for the current design and marks where they diverge. The lumped model's 12 dB per octave fall above the coupled resonance is plotted and labelled as a model artefact, so no one mistakes it for the design.

L3: the companion solver. No commercial finite-element package is assumed or supported. The project builds its own three-dimensional solver as a service from open-source modules, and the browser talks to it directly, so nothing is ever exported by hand into another application. The service is a Python process on the same machine or on a more capable workstation on the local network, exposing an HTTP and WebSocket interface; the browser sends the parametric geometry, the netlist's element and material data and the frequency grid, and receives N-port matrices, port transfer functions and field slices back into the same project. Stack: gmsh for meshing from the parametric geometry, scikit-fem or FEniCSx for Helmholtz finite-element assembly with the boundary-layer wall condition and equivalent-fluid porous regions, PETSc with MUMPS for the complex sparse solves, and Bempp-cl for exterior radiation and pinna scattering by boundary elements coupled at the ports. All are free and open source.

The reason to build it for this one purpose is that boundary conditions are generated, never typed. The browser already knows what every surface is: rigid cup wall, diaphragm with its velocity shape from Section 5, liner with impedance from the material database, leak slit with its thermoviscous impedance, canal port with the ear load, vent with its radiation load, exterior with an absorbing layer. The service assigns conditions from that description, so the failure mode of general-purpose packages, a plausible-looking field produced by one wrong condition, is designed out. Every run is also checked automatically: reciprocity and passivity of the returned matrix, energy balance, mesh-refinement change at the largest resonances, and the continuity test of Section 2, which requires agreement with the L1 result below the lumped 3 percent frequency within 0.1 dB. A run that fails comes back as failed with the offending check named, never as a curve.

Capacity and cost on the target workstation, a 16-core desktop processor with 96 GB of memory and an AMD graphics card: a cup with pinna and canal at 2 mm quadratic elements is about 85 thousand complex degrees of freedom to 20 kHz and needs one to a few gigabytes per frequency, so a 100-frequency sweep is minutes with several frequencies solved in parallel, and finer meshes to about 300 thousand degrees of freedom fit in memory. The vendor example that needed 28 to 100 GB is within reach of this machine. The boundary-element library can use the graphics card through OpenCL where drivers allow; it is optional and falls back to the processor. Results are cached by geometry hash so parameter changes that leave geometry untouched reuse them, and jobs run asynchronously with progress and cancel in the browser. Above roughly 5 kHz the pinna and inter-subject variability dominate, so even this tier reports bands rather than lines there unless the pinna-panel objective of Section 12 is used; that is physics, not a tool limit. In-browser three-dimensional solving stays limited to the coarse preview of 3 mm spacing, 8 kHz and 100 thousand degrees of freedom with a 1.5 GB memory guard. The exchange contract remains as defined, a complex N-port impedance matrix on the shared 48-points-per-octave log grid with optional drum and entrance transfer functions and provenance metadata, so a result from any other tool could also be imported if one ever becomes available.

Gas properties. Temperature is a parameter from 20 to 37 °C. Speed of sound, density, viscosity and thermal conductivity are recomputed from it, which shifts every mode frequency by about 2 percent across that range and changes boundary-layer thicknesses accordingly.

10. Outputs and analyses

Any node pressure or branch flow in the netlist is a probe, and every derived quantity below is computed from probes, so adding an element automatically extends what can be plotted. Each output carries the validity band of the level it came from and the drive, load and reference point it assumes.

Frequency-domain outputs

Output	Definition	Notes
Pressure at drum reference point	Probe at the ear load's microphone plane	Raw; compensated against a target; or field-related through ISO 11904-2 manikin tables
Pressure at canal entrance, open and blocked, and their ratio	Probes at the canal port with and without the canal attached	The ratio is the pressure division ratio of Section 7
Surface-averaged diaphragm pressure	Front minus rear pressure over the piston	The acoustic load the driver actually sees
Electrical impedance	Voltage over current at the source, magnitude and phase	Readouts: free-air and in-situ resonance, mechanical, electrical and total Q, peak height, nominal impedance at 1 kHz, minimum impedance and its frequency, rated-impedance check
Coil current and dissipated power	From the source branch	Against the coil rating
Displacement, velocity, acceleration	Mechanical node	Against the driver's linear excursion limit
Volume velocity and particle velocity per branch	Branch flow; particle velocity is flow over aperture area	Particle velocity above about 1 m/s in a leak or vent flags the laminar assumption and switches the nonlinear orifice model on in time-domain runs

Output	Definition	Notes
Phase, excess phase, group delay	From the complex response and its minimum-phase counterpart	Group delay from the rational fit, not from wrapped phase on a coarse grid
Poles, zeros and Q table	From the vector fit of the linear network	Names each resonance and the elements that set it

Automated readouts. Coupled resonance frequency; bass extension as the frequency where the drum response is 3 dB below its 500 Hz level, stated with that reference; sensitivity in dB per milliwatt and dB per volt at 500 Hz and 1 kHz; error statistics against the selected target on a twelfth-octave grid normalised at 500 Hz, namely standard deviation, absolute slope and RMS error over 20 Hz to 10 kHz; and the predicted preference score of Section 11 when its conditions are met.

Time-domain outputs. Impulse and step response, energy-time curve, burst decay and a cumulative spectral decay plot that shows which resonances ring, all from the same rational model as the group delay so time and frequency views agree by construction. The full complex response is always kept; excess phase relative to the minimum-phase response is computed and reported rather than assumed zero, because vented and bridged topologies are not guaranteed minimum phase. Drum impulse responses export in the AES69-2022 SOFA convention for headphone responses and as float WAV.

Passive isolation. The same network is driven from the outside: a diffuse ambient pressure source is applied at every ambient node, that is the leak, the rear vent, the porous pad path and a cup-wall transmission element modelled as the shell's mass and compliance. The result is insertion loss at the drum versus frequency, reported per ear following the conventions of ETSI TS 103 640. Low-frequency isolation is leak-limited, midrange isolation follows the cup mass law and the pad's porous path, and the ceiling is the fixture's own self insertion loss, which for a GRAS 45CA exceeds 50 dB from 80 to 250 Hz and 65 dB from 350 Hz to 4 kHz per the manufacturer's data. An occlusion-effect estimate drives the canal wall as a volume-velocity source and reports the gain of the occluded canal relative to the open one.

Bleed. The volume velocity leaving through the rear grille and the pad leak radiates into a radiation impedance; the pressure at 0.3 m and 1 m is shown as a band of plus or minus 6 dB with the dipole correction of Appendix C9 below the frequency where the front-to-

rear separation is a wavelength over 2π , and a baffled-piston directivity factor above 2 kHz.

ANC plant paths. Microphones are netlist probes: a feedback or error microphone inside the front cavity and one or more feedforward reference microphones on the cup exterior. The network becomes a two-input, N-output transfer matrix from driver voltage and ambient blocked pressure entering through leak, pad, shell and vents, to the microphones, the drum and the diaphragm. From it the tool computes the secondary path from driver to error microphone and to the drum, the primary paths from ambient to error microphone and to the drum, the reference paths, the driver-to-reference-microphone feedback paths that limit hear-through gain, and the error-microphone-to-drum transfer that decides how much cancellation at the microphone reaches the ear. Each is exported with true phase and explicit delays as passivity-enforced state-space, transfer functions and sampled responses, together with an uncertainty set over fit states and handling cases such as a hand over the cup. A delay budget stacks acoustic standoff at $29\text{ }\mu\text{s}$ per 10 mm, converter and DSP latency from a back-end preset, and driver phase lag, and reports the maximum gain-crossover for a chosen phase margin. Reference points from a customised commercial over-ear feedback design: an insertion-gain crossover of at least 1.5 kHz with a fixed filter and about 2 kHz after in-situ customisation at about 45 degrees of phase margin, sensitivity peaks near 5 dB in good designs, and peaks of 10 to 15 dB treated as a sign of impending oscillation; academic robust designs cap the peak at 3 dB. The Bode sensitivity integral is plotted with its positive and negative areas so the mid-band bump that pays for bass attenuation is visible. The lumped tier's phase is untrustworthy exactly where the crossover sits, so stability conclusions require an L1 or measured plant and the panel enforces that. Results are reported in IEC 60268-24 terms with the open ear on the same head as reference, passive, active and total insertion loss under pink noise and the standard's aircraft, train and cafeteria spectra, and in ETSI TS 103 640 terms for leak robustness. Controller design happens elsewhere; the feedback panel is a phase-2 deliverable and adaptive simulation an optional worker job.

Distortion. With the nonlinear driver model of Section 5 active, the time-domain engine reports total harmonic distortion versus frequency at the characteristic voltage and 6 dB either side, intermodulation with a 70 Hz tone 12 dB above a swept upper tone, multitone distortion, peak excursion and DC offset, and the linear excursion limit by the 10 percent distortion criterion. Leak and vent orifices use a flow-dependent resistance. Results flag test frequencies whose harmonics hit the coupler's half-wave resonance. The readout states plainly that excursion in a sealed cup at 100 dB is a few micrometres, so motor and suspension nonlinearity are usually small and inductance modulation, leak flow and diaphragm buckling dominate measured distortion.

Sensitivity and robustness. Adjoint Jacobians give a heat map of decibels per percent change over frequency and parameter, and a tornado chart at the pinned frequency or for any band metric. Monte Carlo over fit variation from Section 8, ear variation from Section 7 and manufacturing tolerance from Section 12 gives median and 10 to 90 percent envelopes for any output. Design-of-experiments tables list every run with its metrics and netlist hash.

DSP co-design. For active products the acoustic brief is usually to make the response correctable within a headroom budget, so the tool reports the maximum equalisation boost needed to reach a target and the Q of the dips it must fill, a minimum-phase check on the correction, the excursion, coil power and amplifier voltage the corrected response demands at a stated level under the IEC 60268-24 aircraft and train spectra, and how the required correction moves across the fit Monte Carlo, since a fixed equaliser cannot track a secondary path that changes by up to 30 dB between sealed and loose fits in the 80 to 200 Hz band. Resulting filters export as parametric equaliser text in the Equalizer APO and AutoEq convention.

Regulatory and safety panel. With the drive convention stated, the tool reports the A-weighted level at the drum for programme-simulation noise against EN 50332-1 and -2 of 2013 and -3 of 2017, including the 85 dB(A) warning and 100 dB(A) maximum rules and the 150 mV and 75 mV player and headphone figures, IEC 62368-1:2023 clause 10.6, and ITU-T H.870 (03/2022) safe-listening dose management, using IEC 61672-1 weighting. The panel is marked advisory and quotes clause text only after the text has been read. Hearing-protector standards apply only if headsets enter scope.

Comparisons. Any two or more designs or measurements overlay with difference curves, band-averaged deltas and the sidecar compatibility check of Section 14. The frozen-baseline and crosshair behaviour of v1 is kept.

11. Targets, scoring and preference models

A target is a typed object that names its fixture, and the tool never draws one without it. Every published target is a drum-reference-point curve on a specific fixture, and moving a curve to a different fixture changes it by several decibels above 2 kHz and in the bass for inserts. Olive and Clark's 2025 work exists precisely because the Harman target was being misapplied to fixtures it was not derived on.

Target object. Name, version, fixture, reference point, baseline, normalisation frequency of 500 Hz, valid range, provenance class of peer-reviewed, manufacturer or community, and licence. Copyrighted data tables are not embedded. Where a target is legitimately

parametric, the tool ships the reconstruction, a fixture diffuse-field response plus published shelf parameters, flagged as an approximation, and imports user-supplied curves otherwise.

Target	Fixture it belongs to	Status
Harman around-ear and on-ear, 2013, 2015, 2018	GRAS 45CA-class fixture with an IEC 60318-1 ear simulator and Harman's custom leakage pinna, per Olive 2022; the exact coupler configuration is to be confirmed from Olive et al. 2018	Peer-reviewed; data not redistributable; parametric approximation shipped
Harman in-ear, 2017 and 2019	IEC 60318-4 simulator with canal extender	Peer-reviewed; same handling
Diffuse field and free field	Fixture-specific manikin responses per ISO 11904-2; origin Theile 1986	Standards; consistently least preferred in blind tests
Over-ear target for the B&K 5128	Type 4.3	Ravizza et al. 2023; curves and ratings CC-BY-4.0 on Zenodo, shippable
Fixture-specific Harman-equivalent targets	45CA-10 with KB5000 pinnae and 5128	Olive and Clark 2025; data availability to be confirmed
Community targets built on tilted diffuse field	5128 and 60318-4 clones	Community; not peer-reviewed; imported, never bundled
High-frequency extension for inserts	Damped high-frequency 60318-4	Miller and Downey, JAES 2023: a 16 kHz peak of Q 4 at plus 12 dB for young listeners, rising 3 dB per decade of hearing age to plus 21 dB; offered with a plus or minus 5 dB user range and only with damped couplers or the 5128

Fixture translation. A target moves to another fixture by adding the mean difference of several headphones measured on both, and the spread of that difference across headphones is displayed because the interaction with the pinna makes it several decibels above 2 kHz. Default deltas ship only where derived from published paired data.

Preference score. The Harman model is implemented exactly as specified: left-right average, twelfth-octave grid from 20 Hz to 20 kHz, normalised at 500 Hz, standard deviation and absolute slope of the error over 20 Hz to 10 kHz, mean absolute error over 40 Hz to 10 kHz. Around-ear and on-ear: 114.49 minus 12.62 times standard deviation minus 15.52 times slope, correlation 0.86 and error 6.7 points per Olive 2022. In-ear: 68.685 minus 3.238 times standard deviation minus 4.473 times slope minus 2.658 times mean error, correlation 0.91. The score is greyed out whenever the fixture and target pair differ from the training pair, when applied to simulated rather than measured curves the interface says so, and the 20 to 100 Hz band is flagged as coupler-extrapolated for 60318-4 loads, which the standard does not validate below 100 Hz.

Error metric set. Never one number: RMS error, standard deviation, absolute slope, mean error, band-limited RMS error for 20 to 200 Hz, 200 Hz to 2 kHz, 2 to 8 kHz and 8 to 20 kHz, tolerance-mask compliance in percent, and the ITU-R BS.708 mask, which is plus or minus 2 dB below about 250 Hz, plus or minus 1.5 dB from 500 Hz to 4 kHz, and plus or minus 4 dB at 16 kHz. Every metric states its frequency range; above 10 kHz is reported separately.

Bounds, not lines. Listener segmentation gives three classes: 64 percent prefer the target as is, 15 percent want 4 to 6 dB more bass, 21 percent want 2 dB less, per Olive, Welti and Khonsaripour 2019. Preference bands are drawn from these, widened above 2 kHz by the inter-individual spread of headphone transfer functions, where the main ear resonance varies by about 2 kHz and 10 dB across people, and widened above 8 kHz by fixture validity. Bass and treble shelf controls near 105 Hz and 2.5 kHz are first-class personalisation parameters, stored separately from the reference target.

Level and content. Preferred balance depends on playback level, so every comparison states its reference level, and ISO 226:2023 equal-loudness contours with ISO 532-1 loudness show how tonal balance shifts with level. Spatial and binaural content prefers a different balance from stereo, per Engel et al. 2022, and the tool notes which content class a target was derived with.

Compensation pipeline. Field-related views subtract the fixture's diffuse-field or free-field manikin response in decibels per ISO 11904-2 and IEC 60268-7, and report the effective frequency range of the compensated response, a quantity the 2025 edition of IEC 60268-7 introduces. Level normalisation at 500 Hz, or at the 200 to 500 Hz band mean, and loudness alignment per ITU-R BS.1770 precede every comparison.

Left-right tracking. Unit-to-unit and fit differences between ears are scored against the BS.708 tracking limits of 1 dB from 100 Hz to 8 kHz and 2 dB from 10 to 16 kHz, and the left-right tracking response defined in IEC 60268-7:2025.

Limits stated in the interface. Harman targets rest on panels of at most about 300 mostly trained listeners with limited programme; diffuse-field and free-field targets remain in the standard but rank low in every controlled test; community 5128 targets are not peer-reviewed; no preference model is validated above 10 kHz or on simulated curves. An adaptive preference-elicitation module in the style of Ravizza et al. 2025, paired comparisons over octave-band gains driven by the model-derived audition filter, is specified as a separable module rather than part of the design workbench.

12. Design workflow

The tool is built around the loop a headphone engineer actually runs: concept, parametric model, target fit, robustness check, prototype, measurement, model fit to the measurement, and back. The model-to-measurement round trip is a first-class feature, not an import menu, because a lumped model that has never been reconciled with a measured impedance curve is a sketch.

flowchart LR

```

A[Concept and geometry] --> B[Parametric netlist]
B --> C[Target fit and sensitivity]
C --> D[Fit and tolerance robustness]
D --> E[Prototype]
E --> F[Measure Z and FR]
F --> G[Identify parameters]
G --> B
D --> H[Companion solver]
H --> B

```

Each pass tightens the model; the companion solver is entered only when a candidate survives the robustness check, and its result returns into the same netlist.

Parameter identification from measurements. Import measured impedance and frequency response with their metadata, then fit the netlist to them by weighted least squares on complex impedance and log-magnitude response, 10 Hz to 20 kHz, Levenberg-Marquardt in log-parameter space with a CMA-ES global start when requested. The impedance model includes a lossy-inductance branch and a log-frequency suspension-

creep compliance, because headphone drivers depart from a constant compliance by up to half at low frequency and a pure inductance misses the eddy-current rise. Reported parameters carry confidence intervals from the Jacobian covariance.

Identifiability is enforced, not assumed. Impedance alone determines only the combinations Bl^2/Mms , $Bl^2 \cdot Cms$ and Bl^2/Rms . After an impedance-only fit the tool marks Bl , Mms and Cms as scale-ambiguous and asks for one of: an added mass, a known coupler volume with a measured effective area, a laser displacement transfer function, or a specified SPL in a known load. The singular values of the weighted Jacobian are shown so the user sees which parameters the data constrain. Fitting SPL alone is refused for the same reason. The added-mass method is flagged as unreliable for drivers under about half a gram, where a 10 mg adhesive dot is 3 percent of the moving mass.

Inverse design. The user picks an objective over a frequency band: deviation from a target curve, a preference score, isolation, sensitivity, bass extension, or a weighted mix. The optimiser varies chosen continuous parameters within bounds and discrete choices such as mesh grades from the database. Results are ranked with the sensitivity of the objective to each parameter, so a design that only works at one exact vent area is visible as such. Objectives are restricted to the credible band of the active fidelity level.

Pinna-panel objective. Above about 4 kHz the response at the drum depends on the individual pinna, and a headphone that drives the ear from an unnatural direction makes that dependence a nuisance, while one that drives it like a distant frontal source turns it into the listener's own, expected cues. The companion solver therefore runs a design against a panel of scanned pinnae rather than one fixture ear. For each pinna in the panel it computes two things with the same mesh: the headphone-to-drum transfer with the cup in place, and that pinna's free-field transfer to the drum for a chosen loudspeaker direction such as 30 degrees azimuth. The objective is the mean over the panel of the deviation between the two, up to a smooth common shelf, not the across-panel spread of the headphone response by itself. Design variables exposed to it are driver angle, offset and standoff, cup interior absorption, and multi-driver arrangement. Panel meshes come from public head-and-pinna scan sets such as the HUTUBS collection, subject to licence review, and the same run reports how a pinna-inclusive target curve would over- or under-count the listener's own pinna for that design.

Sensitivity and tolerance. Adjoint Jacobians give decibel-per-percent heat maps across frequency and a tornado chart of the parameters that matter. Tolerance analysis samples datasheet and manufacturing scatter, for example coil resistance within 5 percent and

free-air resonance within 15 percent, together with fit variation from Section 8, by Latin hypercube, and plots median and 10 to 90 percent envelopes. Dimensional tolerances of the ear simulator itself are included, following the Monte Carlo analysis of Luan et al. 2019.

Manufacturing tolerance framework. Each parameter has a tolerance class with a source: driver resonance within 15 percent and coil resistance within 5 percent from the reference datasheet; mesh resistance within 10 to 15 percent as industry practice, since suppliers state none; pad foam density and flow resistivity within 30 percent plus compression; moulded gaps within 0.05 to 0.1 mm as an engineering estimate. Tolerances propagate to process-capability statistics of band levels and to left-right mismatch scored against the ITU-R BS.708 tracking limits and the tracking response of IEC 60268-7:2025. Driver pairing and binning strategies are compared as alternative input distributions, because left-right matching yield is set by which parameters the acoustics amplify.

Measurement uncertainty budget. Every imported and simulated curve carries an uncertainty budget: coupler tolerance from the standard's table, microphone calibration, repositioning spread from three to five placements, fixture-to-human translation, and solver numerical error. The budget sets the fitting weights and the pass-fail margins, so a residual is judged against what the measurement can resolve rather than over-fitted to one placement.

Cost and trade-offs. Components may carry cost fields, and the optimiser can report a Pareto front over bass extension, isolation, leak robustness, excursion margin, left-right yield and cost instead of a single scalar, because design reviews are trade-off discussions and a single preference score hides the choices an engineer has to defend.

Design of experiments. Full and fractional factorial and Latin hypercube sweeps over any parameter set, with response metrics per run: low-frequency corner, standard deviation and slope of the error against target, sensitivity, distortion at the characteristic voltage, isolation. Results export as a table with the netlist hash of every run.

Versioning and comparison. Designs are versioned, diffable at the netlist level, and comparable in overlays with the frozen-baseline behaviour v1 had, extended to any number of curves. A report generator produces a dated summary with the plots, metrics, provenance and the validity band of every curve.

Production hand-off. End-of-line limit curves are derived from the 5 to 95 percent tolerance envelope plus measurement uncertainty for drum response, impedance magnitude with the 80 percent-of-rated check, polarity and left-right tracking, with a golden-unit correlation between the design ear load and the quality fixture, typically a

GRAS 45CC for over-ear per Klippel practice. Limits export as CSV because vendor native formats were not confirmed. A quality-control view shows left-right balance in third octaves below 10 kHz, band average level and a rub-and-buzz placeholder, so R&D output translates directly into a production test sequence.

13. Materials and component database

Every database entry carries its value, unit, measurement method, physical state, tolerance, provenance and licence, and the interface shows whether a number is measured, from a datasheet or estimated. The highest-priority gap found in this work is that no public measurements exist for the materials headphone engineers use most: pad face fabrics, pad foams and headphone-grade fleece. Those entries ship as visibly estimated defaults until a measurement campaign fills them.

Entry metadata. Flow resistivity by ISO 9053-1 static or ISO 9053-2 alternating airflow, or ASTM C522; porous parameters by ISO 10534-2 impedance tube with Panneton-Olmy inversion, or ASTM E1050; compression state and temperature; anisotropy where measured; tolerance where the supplier states one. Saati sheets state no tolerance, and industry practice assumes 10 to 15 percent. Gore states a unit-to-unit standard deviation near 0.2 dB at 1 kHz for one membrane vent.

Meshes and vents. The full Saati Acoustex woven-mesh range, where the grade number equals the specific flow resistance in MKS rayl and each entry carries pore size, open area, thickness and weight: 3 rayl with 285 µm pores and 44 percent open area at one end, 260 rayl with 18 µm pores and 13 percent open area at the other, with grades 006 through 160 between. Gore ePTFE acoustic vents from 45 to 149,000 rayl with thickness and the manufacturer's 1 kHz insertion-loss data. Sefar and Nitto membranes as user-entered values because their per-grade tables were not retrievable.

Porous materials and fabrics

Material	Value	Status
Glass wool, 14 kg/m³	Flow resistivity 5.9 kPa·s/m² through-thickness, 2.9 in-plane	Measured, Tarnow 2002
Mineral wool boards, 60 to 80 kg/m³	15 to 28 kPa·s/m²	Measured, conference source not independently located

Material	Value	Status
Open-cell polyurethane foam, acoustic grade	About 50 kPa·s/m ²	Single measurement
Polyurethane foam, textbook porous set	Porosity 0.85, flow resistivity 34 kPa·s/m ² , tortuosity 1.18, viscous length 60 µm, thermal length 87 µm	Allard and Atalla, reused in COMSOL's headphone model
Polyester fleece	Flow resistivity from bulk density by the Garai-Pompoli and Bies-Hansen regressions	Coefficients to be verified against the papers before hard-coding
Reticulated foam by pores per inch	Ranges only	Unverified; editable defaults with badge
Woven textiles	Transparent facing about 30 rayl; engineered acoustic textiles 200 and 600; wool-felt upholstery 650; cotton-polyester upholstery 2300	ISO 9053 measurements, Hough and Law 2025
Velour pad face, felt, protein leather	Not measured; velour expected 50 to 600 rayl, protein leather impermeable	Estimated; measurement campaign required

Drivers. A curated record set with the governance rules of Section 5: the Tymphany HPD-40N16PET00-32 and HPD-50N25PR00-32 as verified datasheet records, fields for every parameter with measurement condition, automatic identity checks on resonance, electrical Q and equivalent volume, and unit-anomaly flags. Two published parameter sets were found to be physically impossible during this work, one quoting a moving mass in micrograms, so no imported record enters a design unchecked.

Diaphragm materials. Beryllium at 1.85 g/cm³ and 287 to 310 GPa, titanium 4.5 and 120, aluminium 2.7 and 70, magnesium 1.74 and 45, PET 1.38 and about 3 GPa with glass transition 67 to 81 °C, PEEK 1.32 and 3.6 GPa with glass transition 143 °C, bacterial cellulose up to about 15 GPa. Loss factors of films at audio frequencies were not found and are import-only.

Ear loads and fixtures. The IEC 60318-4 network fitted to the standard's table and its damped variant; the P.57 Type 4.3 canal from the standard's public geometry and transfer-impedance target; Type 4.4; IEC 60318-1 only when its table is sourced; IEC 60318-8 as a

further coupler; fixture geometry presets for flat-plate, 45CA, KEMAR, 5128 and HEC heads; user tables.

Signal chain. Amplifier presets by output impedance and clipping; DSP back-end presets by latency class: an analog or very high rate loop near 5 μ s, a 96 kHz codec near 30 to 40 μ s, and 48 kHz block processing at a millisecond or more, the last of which rules out broadband feedback ANC. The 5 μ s figure comes from vendor forum text and is marked unconfirmed until read from a datasheet.

Magnets and wire. Optional records for neodymium grades and copper, copper-clad aluminium and aluminium wire with resistivity, density and temperature coefficient, feeding a coil calculator that keeps resistance, inductance, moving mass and force factor mutually consistent. Predicting force factor versus displacement from magnet geometry needs magnetostatic finite elements and is out of scope; those curves are imported.

Cost fields. Each component may carry a unit cost so the multi-objective optimiser of Section 12 can report cost alongside acoustics. Public anchors: a standard 40 mm headphone driver at roughly 0.60 to 0.70 US dollars in volume and about double for upscale materials, per Klasco and Wong 2019.

Data licensing register

Dataset	Licence	Handling
Ravizza et al. 2023 5128 target curves and ratings	CC-BY-4.0 on Zenodo	Bundled with attribution
Hearpiece individual transfer-function database, Denk and Kollmeier 2020	Public on Zenodo	Bundled for validation of in-ear variants only
PANDAR in-ear ANC path database	Terms unconfirmed	Import by the user
AutoEq	Code MIT; measurement data owned by each source	Import format supported; data not bundled
squig.link and CrinGraph	Code 0BSD; data owned by contributors	Import format supported; data not bundled
oratory1990, Crinacle, Rtings, Audio Science Review	Not redistributable	Manual paste or upload with provenance tag

Dataset	Licence	Handling
IEC, ITU-T and ISO tables	Copyrighted	Fitted models and citations only, never verbatim tables
Saati and Gore datasheet values	Manufacturer data	Cited per entry
Klippel and Listen native limit formats	Unconfirmed	CSV export instead

14. Import, export and interoperability

Every curve that enters or leaves the tool carries a metadata sidecar, and the tool refuses to overlay curves whose fixture, compensation or drive convention differ without an explicit override. Most published headphone comparisons are wrong because of exactly this mismatch.

Metadata sidecar. JSON with fixture and ear simulator, pinna type, number of seatings and averaging method, smoothing, drive convention and level, source impedance, compensation state, reference point, temperature, date, device identifier and provenance. Magnitude averaging and complex averaging of seatings give different treble results, so the method is stored.

Direction	Content	Formats
Import	Impedance versus frequency	FRD, ZMA, TXT, CSV; Klippel and DATS exports through adapters
Import	Frequency response at a stated reference point	FRD, TXT, CSV following REW, CrinGraph and AutoEq conventions; WAV impulse responses
Import	Target curves	CSV with fixture tag
Import	Driver parameters	Thiele-Small sets, Klippel large-signal polynomial tables, reduced-order state-space driver models
Import	Distributed models	Complex N-port impedance matrices on a log grid from the companion solver or any other tool, with port metadata

Direction	Content	Formats
Import	Material data	Flow resistivity, porosity, tortuosity and characteristic lengths per ISO 9053 style measurement
Export	Netlist	Versioned JSON project schema in SI units; SPICE-compatible netlist for cross-checking in circuit simulators
Export	Curves	CSV of any probe with sidecar; PNG and SVG plots
Export	Impulse responses	32-bit float WAV at 48 or 96 kHz, minimum or mixed phase, with the audition state
Export	Plant models	Descriptor state-space and transfer functions from vector fitting with passivity enforced, for controller design
Export	Geometry and boundary data	Parametric geometry with named port faces, driver velocity spectrum, port and liner impedance tables, ear termination, sent automatically to the companion solver of Section 9 and readable by other tools if ever needed
Export	Reports	Dated summary with plots, metrics, provenance and validity bands

Further formats. Drum impulse responses export in the AES69-2022 SOFA convention for headphone responses so binaural and virtual-reality teams can consume them directly. Correction filters export as parametric equaliser text in the Equalizer APO and AutoEq convention and as FIR WAV for convolvers. Measurement exports follow the REW text convention so they round-trip through common tools.

Conventions. Frequency grids are log-spaced at 48 points per octave unless the source states otherwise. Impedance matrices are in $\text{Pa}\cdot\text{s}/\text{m}^3$ with positive flow into the domain. Sensitivity is exported in both dB SPL per volt and dB SPL per milliwatt with the reference frequency and coupler stated, following IEC 60268-7. Unit slips are common in datasheets, so imported Thiele-Small sets are cross-checked against the identity between resonance, mass and compliance before use.

Project model, versioning and privacy. The project is schema-versioned JSON with semantic versioning and migrations, and a text netlist is the canonical, git-friendly artefact. Designs branch into variants with side-by-side diffs of parameters and curves and an undo history. Every export embeds a reproducibility hash covering solver version, air constants, frequency grid and element-library version. Storage is offline-first in IndexedDB or the origin private file system with persistent storage requested, plus File System Access save and open on browsers that have it and a download fallback elsewhere; Safari evicts idle site data after seven days, so the tool warns and offers export. No design data leaves the browser unless the user shares a project, and an encrypted project export is offered.

Public measurement databases. Community measurement sets are useful for target-metric regression and for orientation, not for physics validation, and each has its own rig and licence. The tool bundles none of them; it imports from them with the sidecar filled in by the user, and documents which fixture each well-known source uses.

15. User interface

The interface is three synchronised views of one netlist: a dimensioned cross-section, an auto-laid-out schematic, and the measurement plots. Change a value in any view and the other two update from the same solve. Nothing on screen is hand-drawn to look plausible; every shape, value and explanatory sentence derives from the netlist and the solver.

Parametric cross-section. Geometry is entered in millimetres: pad inner opening and depth, pinna and concha volume, driver position and diameter, baffle thickness and vent layout, cup radius and depth, grille pattern, liner thickness. Cavity volumes, port lengths and areas, and slit dimensions are computed from this geometry and pushed into the netlist, so the picture and the physics cannot disagree. The ear model is drawn to scale with its canal length and reference plane. Fixture surface (flat plate or head-shaped) is a selectable backdrop because it changes the pad seal model.

Netlist-driven schematic. The schematic is generated from the netlist in three lanes for the electrical, mechanical and acoustic domains, with couplers drawn between lanes. Each element shows its value, its fidelity representation and a colour for its own validity: green while its k_L is under 0.5, amber to 1.0, red beyond, the same 10 and 36 percent error thresholds the plot shading uses. Hovering a slider glows the element it drives, as in v1; clicking an element opens its model card with equations, parameters, provenance and derivative of the active output with respect to it.

Plots. All plots share a log frequency axis from 10 Hz to 40 kHz with 20 Hz to 20 kHz as default view, validity shading from Section 2, a crosshair that reports every visible curve and the differences between them, a pinned inspection frequency, unlimited frozen overlays, and labelled smoothing from none to third-octave. Available plots:

- Pressure at drum reference point and canal entrance, raw and compensated against a target, with population and fit envelopes
- Electrical impedance magnitude and phase with the Thiele-Small readouts
- Displacement, velocity and volume velocity of any branch
- Phase, group delay, impulse and step response
- Isolation, bleed and the ANC plant paths
- Distortion versus frequency and level
- Sensitivity heat map and tornado chart
- Lumped versus modal versus finite-element overlay

Field and mode views. Cavity mode shapes as coloured slices of the modal or finite-element solution at the pinned frequency, diaphragm mode shapes, and the pressure field in the axisymmetric model, so a user sees where a standing wave lives before choosing where to put a liner.

Particle animation. Retained from v1 and corrected: particles oscillate rather than drift, amplitude proportional to particle velocity, which is volume velocity divided by aperture area, phase taken from the solution so port flow visibly flips relative to the diaphragm through resonance. Each path has a phasor dial beside it. The animation runs at display rate from the last completed solve and never blocks it.

Explain panel. Sentences of the form "raising front volume 10 percent lowers drum pressure 1.8 dB from 40 to 300 Hz and moves the coupled resonance from 1.20 to 1.14 kHz" are generated from the adjoint sensitivities and the solved curves, ranked by effect size, and restricted to the credible band. No cause-and-effect text is hand-written anywhere in the interface, because v1's hand-written vent description was backwards.

Presets and case studies. Each preset opens with a what-to-look-for card listing the two or three features the design is meant to demonstrate, each linked to the plot region and the elements responsible, and with its provenance fields visible.

Responsiveness. L0 recomputes on every input event, coalesced to one solve per animation frame in a worker. Higher levels show the previous result as a ghost curve until their recompute lands. Long jobs such as Monte Carlo, optimisation and finite-element sweeps run with progress and cancel controls.

Conventions panel. Active drive convention, source impedance, ear load, temperature, reference point, smoothing and compensation are shown in one persistent strip on every plot, and are written into every export.

Accessibility and working practice. WCAG 2.2 Level AA: colour is never the only carrier of meaning, text contrast of at least 4.5 to 1 and non-text contrast of 3 to 1, plot series distinguished by pattern as well as hue, full keyboard operation of the netlist and schematic editors with numeric entry on every slider, a screen-reader-accessible data table behind every curve, a text alternative for the animation, and safe audition-level defaults. A units toggle never mixes systems. The interface works at laptop width; the tool is not designed for phones.

16. Auralization

The audition path is the model's own transfer function convolved with the programme, level matched, and by default it plays the difference between two designs rather than an absolute response. The listener's own headphones multiply everything they hear, so an absolute rendering of the simulated headphone is not something anyone can hear; a difference between candidate and baseline is. This is the same logic as Harman's virtual headphone method, in which one replicator headphone equalised to each candidate's measured magnitude response reproduced the preference ratings of the real headphones.

What is auralized. The filter is the candidate response divided by a baseline at the same reference point. The baseline can be a saved design, a target curve, or an imported measurement. An uncompensated absolute mode exists as an explicitly flagged diagnostic and nothing more. Every audition state is serialisable: design identifiers, phase mode, level-match method and gain, programme file hash and noise seed, so listening notes can be reproduced.

From response to impulse response. The network is re-solved directly on the linear FFT grid used for the filter, 8192 points at 48 kHz by default and selectable from 4096 to 16384, with the length sized to cover the decay of the lowest-frequency high-Q pole, about 2.2 times Q over frequency. It is never built by interpolating real and imaginary parts of the sparse log-spaced plotting grid, which produces ripple and non-causal artefacts. Outside the solved band the response is extrapolated with the model's own asymptotic slopes and tapered. The full complex response is always computed and its excess phase estimated against the minimum-phase version from the real cepstrum. Minimum phase is the default only when the excess group delay within the trusted band stays under about 0.5 ms, a

threshold set below the one to two millisecond group-delay audibility figures usually attributed to Blauert and Laws 1978, to be confirmed from that paper before release; otherwise the mixed-phase response is used and its pure delay reported, because vented and bridged topologies are not guaranteed minimum phase. Three phase modes are offered:

1. Minimum phase by the real cepstrum with generous zero padding. Default for difference filters when the excess-phase test passes, and forced above the model's declared validity frequency.
2. Mixed phase from the model's complex response, with delay alignment, Hermitian symmetry enforced including real DC and Nyquist bins, a Tukey window and a guard band.
3. Linear phase, diagnostic only, because its pre-ringing would be blamed on the design.

Convolution engine. An AudioWorklet runs uniformly partitioned frequency-domain convolution with the block equal to the render quantum and FFT size twice that, preallocated buffers, a WebAssembly SIMD FFT, and independent left and right filters; a 16384-tap filter at 128-sample blocks costs about 50 million floating-point operations per second per channel, a few percent of one core. Filter updates crossfade linearly over one or two blocks so slider drags do not click. Nothing allocates or posts messages inside the audio callback. The render quantum is 128 frames everywhere; the configurable hint is Chromium-only and treated as an optimisation. The native ConvolverNode is a fallback for non-isolated contexts, with normalisation disabled before the buffer is set, since the default normalisation destroys level matching; it rejects a buffer at a different sample rate with an exception rather than resampling, and only WebKit moves long filters to a background thread, so its filter-length cap applies to Safari alone. The audio context is requested at 48 kHz so the loudness and pink-noise coefficients apply verbatim; if the device refuses, the filters are redesigned for the actual rate.

Level matching and safety. Every A/B pair is matched by integrated loudness per ITU-R BS.1770-5, computed over the actual programme after convolution, to within 0.1 LU. Loudness and true-peak meters are shown. A-weighted RMS and a mid-band anchor are offered as alternatives, and the interface always states which is active. Pure tones fall outside the loudness algorithm's scope and are matched by RMS. A hard true-peak limiter sits at the output and the start-up gain defaults low, because a 12 dB correction applied to pink noise can otherwise reach a harmful level in the listener's own headphones.

Programme material. Seeded pink noise by the Kellet filter, refitted for 48 and 96 kHz rather than reusing its 44.1 kHz coefficients, and by the Voss-McCartney method; white noise; a sine sweep; and user files decoded by the browser with a length cap or a

streaming decoder in a worker. Pink noise is the most revealing signal for short-delay resonances, and low-Q deviations of a few tenths of a decibel are audible on it, so the convolved chain must reproduce the analytic response within 0.1 dB from 20 Hz to 20 kHz.

Two ears. Left and right carry separate parameter vectors. Monte Carlo fit draws are pushed to both ears independently, an interaural level difference readout in third-octave bands flags differences above about 1 dB, and an interaural phase readout covers frequencies below 1.5 kHz.

Inverting measured responses. When a target or a measured response is inverted for the baseline, the inversion uses Kirkeby-Nelson frequency-dependent regularisation with sixth-octave pre-smoothing, a boost cap of 12 dB, no inversion of notches deeper than 15 dB, and a band limit of about 20 Hz to 10 kHz.

Diagnostics. A panel reports the context sample rate, base and output latency where the browser exposes them, render quantum, cross-origin isolation state, SIMD availability, and measured solve and filter-synthesis times. The tool promises no fixed latency figure because browsers do not expose a reliable one.

Optional listening-test module. An ABX or multi-stimulus trial logger with hidden reference, for internal listening sessions, storing the serialised audition state with each trial.

17. Validation and test suite

A build ships only when it passes three layers of tests: analytical derivations with numeric tolerances, network property tests run on every solve, and regression against reference data. The analytical layer is normative because it catches the class of error v1 had: a model whose copy claimed effects its equations could not produce.

Analytical checks. Each is a unit test against a closed-form result, using air at 20 °C: density 1.204 kg/m³, speed of sound 343 m/s, viscosity 1.81e-5 Pa·s, ratio of specific heats 1.4. Derivations are in the appendix.

Check	Setup	Pass criterion
Pressure-chamber flatness	Sealed front cavity, stiffness-controlled diaphragm, constant voltage	SPL flat within 0.1 dB from 20 Hz to one tenth of the coupled resonance; level within 0.05 dB of $p c^2 \cdot S d \cdot x / V$. Example: 10 cm², 30 cm³, 1 µm gives 107.5 dB

Check	Setup	Pass criterion
Coupled resonance and high-frequency slope	Same, lossless, swept to 16 kHz	Resonance within 0.5 percent of the formula; slope of minus 12 dB per octave within 0.3 dB over 4 to 16 kHz. Example: 0.1 g, 1 mm/N, 10 cm ² , 30 cm ³ gives 1204 Hz
Leak slopes	Cavity shunted by a pure resistance, then a pure inertance, then both	Plus 6 and plus 12 dB per octave within 0.3 dB over two octaves below the respective corner; mixed leak shows the transition and a Helmholtz peak with Q within 5 percent, with its resistive-to-inertive corner placed at least a decade below the test band
Slit leak worked examples	1 mm by 30 mm by 10 mm slit; 0.2 mm by 30 mm by 10 mm slit	The first is inertive above a few tens of hertz and resonates with 30 cm ³ near 550 Hz with a Q far below the Poiseuille estimate, because its resistance grows with the square root of frequency; the second is resistive with a 6 dB per octave corner at 83 Hz and no resonance
Helmholtz vent	100 cm ³ cup, one 3 mm hole in a 2 mm wall	215 Hz with both-side flanged end corrections; a deliberate no-correction case must fail by more than 30 percent

Check	Setup	Pass criterion
Impedance peak	Free-air driver	Peak height equals Re plus Bl^2/Rms within 0.1 percent; adding a sealed cavity moves the peak so that the squared frequencies differ by $Sd^2/(4\pi^2 \cdot Mms \cdot Caf)$ within 0.5 percent; adding acoustic resistance lowers the peak as predicted
Fill thermodynamics	Cavity fully filled, then 25 percent	SPL falls 2.92 dB at full fill and 0.83 dB at 25 percent in the low-frequency limit; gain vanishes above the thermal relaxation frequency
Sensitivity scaling	Coil rewound from 32 to 300 Ω at constant mass	Minus 9.72 dB at constant voltage, zero at constant power, plus 9.72 dB at constant current
Lumped validity	Lumped compliance against a transmission-line cavity	Agree within 0.3 dB below kL of 0.3; diverge by about 3 dB at kL of 1
Ear simulator	IEC 60318-4 network	Table 1 transfer-impedance levels at all 21 frequencies within tolerance; 1260 mm ³ at 500 Hz; half-wave at 13.5 kHz plus or minus 1.5 kHz
Thermoviscous limits	Slit and tube elements	Low-frequency limits recover 12 $\mu l / (wh^3)$ and 6 $\rho l / (5wh)$ for slits, 8 $\mu L / (\pi a^4)$ and 4 $\rho L / (3\pi a^2)$ for tubes, to 0.1 percent; high-frequency resistance grows as the square root of frequency
Baffle vent sign	Front-to-rear duct	Increasing its area lowers low-frequency SPL at the drum

Check	Setup	Pass criterion
Bleed	Open-back, rear volume velocity into radiation impedance	Within 6 dB of the monopole-with-dipole-correction estimate

Property tests on every solve

- **Passivity:** the real part of electrical input impedance is at least R_e at every frequency; the real part of every acoustic port impedance is non-negative; every element value is positive. A thermoviscous element with negative resistance means the wrong square-root branch was taken, a bug observed while deriving this suite.
- **Energy balance:** with peak phasors, half the real part of voltage times conjugate current equals half the sum of coil loss including the lossy-inductance real part, mechanical loss and every acoustic resistive dissipation, plus radiated power; with RMS phasors the halves drop; relative error under $1e-9$.
- **Reciprocity:** the acoustic sub-network impedance matrix is symmetric within $1e-9$; the BL gyrator is anti-reciprocal.
- **Two-ports:** determinant of every transfer matrix equals one for reciprocal elements.
- **Causality:** the impulse response from the full-band complex response has pre-response energy below minus 80 dB relative to the main response.
- **Causality and excess phase:** the full complex response satisfies Hilbert-transform consistency; minimum phase is asserted only after checking the zeros of the rational fit, and the excess phase is reported rather than assumed zero.
- **Asymptotes:** pressure over volume velocity tends to a constant or zero as frequency tends to zero according to leak type, and falls 12 dB per octave for the sealed lumped case.
- **Sign convention:** positive source voltage at zero frequency yields positive force toward the ear and positive drum pressure.
- **Dimensional analysis:** every element declares units and the assembler rejects mismatches.

Numerical and DSP tests

- Special functions to $1e-9$ against reference values.
- Thiele-Small round trip: fit f_s , Q_{ms} , Q_{es} and R_e from the simulated impedance and recover the inputs within 1 percent.

- Synthetic fitting: a generated impedance with known creep and lossy-inductance parameters is recovered within 2 percent.
- Impulse response chain: Hermitian realness, Parseval, minimum-phase energy concentration, partitioned convolution equal to direct convolution, magnitude error of the convolved chain at most 0.1 dB from 20 Hz to 20 kHz.
- Loudness on the EBU Tech 3341 reference set; pink noise slope of minus 3 dB per octave within 0.3 dB.
- Golden files per preset for the complex response and impulse response with tolerance-based comparison, since bit-exact results differ across engines. Rust tests run natively and under wasm32.

Reference-data regression

- Driver datasheet sets from at least two 40 and 50 mm headphone drivers: simulated impedance reproduces datasheet resonance and Q values within stated tolerance.
- Fixture curves: the 60318-4 model against the standard's table; the damped variant against the manufacturer's published response.
- Public frequency-response databases are used only for target-metric regression, never for physics validation, and each source's licence is checked before bundling.
- Every named preset stores which parameters are datasheet values and which are estimates, and the regression compares only the former.

Open reference headphone. The normative validation artefact is a 3-D printable circumaural cup around the Tympany HPD-40N16PET00-32 datasheet driver with a defined pad, mesh and vent set, measured on an IEC 60318-4 simulator in undamped and damped form and on a Type 4.3 fixture with at least five re-seats each, plus impedance on the fixture and in free air, published under CC-BY with its geometry. Acceptance after fitting only leak and front volume: within 2 dB to 1 kHz and within 4 dB from 1 to 4 kHz, reported without a bound above. No public dataset combines geometry, impedance and coupler response for one headphone; without this artefact the suite is limited to analytical oracles and the accuracy claim cannot be falsified.

First-release thresholds. The 60318-4 network passes the standard's table at all 21 frequencies. The reference driver's datasheet impedance is reproduced within 3 percent in resonance, minimum impedance and total Q. The reference headphone meets the bounds above. The audition chain is within 0.1 dB of the analytic response in an offline audio context. A 64-node, 512-point solve completes in under 50 ms, stated as a target until the benchmark milestone confirms it.

Validation under limited fixture access. Fixture time is booked and scarce, so the validation campaign is a scripted protocol run by the tool: it lists the measurements a session must produce, pre-computes every prediction, imports the analyser's exports directly, and shows residuals against the uncertainty budget before the session ends, so a missing or failed measurement is caught while the fixture is still available. The clone coupler, once calibrated against the head-and-torso session, carries day-to-day regression at the bench.

18. Presets and case studies

Presets are generic design classes, not product names. Where a commercial product is referenced, the record stores only its public datasheet values as anchors and marks every other parameter as an estimate, and the tool never shows a measured-looking curve for a named product it has not measured. v1's HD600 and M50x presets were internals guessed to fit a name; v2 refuses that.

Presets

Preset	Anchors and construction	What to look for
Reference open 40 mm dynamic	The validation artefact of Section 17: a 3-D printable cup around the Tympany HPD-40N16PET00-32 datasheet driver with defined pad, mesh and vents	Every curve here has measured data behind it; use it to learn how far each level can be trusted
Open dynamic, 300 Ω class	Anchors from a well-known open-back: 300 Ω , 97 dB per volt at 1 kHz by the manufacturer, while an independent measurement gives about 103 dB per volt, which shows that sensitivity depends on coupler, frequency and averaging; impedance peak near 100 Hz rising to roughly 500 Ω per a published sheet; velour pad porous branch; open grille; light rear fleece	Low leak sensitivity because the driver is nearly a pressure source; the bass bump on a 120 Ω source; high bleed

Preset	Anchors and construction	What to look for
Closed dynamic, 38 Ω class	Sealed cup with fleece, protein-leather pad, baffle vent under damping paper; the 38 Ω and 99 dB per milliwatt anchors are from memory and flagged unverified	High leak sensitivity; the baffle-vent sign; cup modes above 3 kHz that only the L1 modal model or L2 shows; isolation
Semi-open aperiodic	Rear vent of 2 mm holes in a 1.5 mm wall on 60 cm ³ , with and without a 160 rayl mesh	The same hole is a 230 Hz resonator with Q near 13 bare and a resistive leak meshed; mesh choice, not geometry, decides
Planar open, 20 to 32 Ω class	Experimental plug-in with public anchors of 20 Ω and 103 dB per milliwatt for one commercial planar and 32 Ω for another	Nearly resistive impedance; air-load-dominated stiffness; many low-Q modes
ANC closed	Closed dynamic plus a feedback microphone node, two feedforward microphone nodes and a back-end latency preset	The delay budget; how front volume and leak move the secondary path; the 1.5 to 2.2 kHz crossover region where lumped phase is untrustworthy
Supra-aural	Pad on pinna, leak-dominated front chamber, IEC 60318-1 coupling, P.57 Type 3.2 low- and high-leak slits	Bass set almost entirely by leak

Modifications as states, not presets. The DIY mods of v1 become fit states or element changes applied to any preset. A front damping mod adds a mesh over the diaphragm and shows the low-frequency effect honestly while stating that its treble effect needs D3 modal data. A taped baffle vent removes the duct and runs the sign test. Eyeglasses and worn pads are the fit states of Section 8, labelled estimates.

What-to-look-for cards. Each preset opens with two or three features, each linked to the plot region and the elements responsible, generated from the solve and the sensitivity ranking rather than written by hand. Expected behaviours from public data, such as the impedance peak of the 300 Ω class, are attached as anchors with their source.

Guided case studies

1. Tune a rear vent to reach a bass target within excursion and isolation constraints, then check it across the fit Monte Carlo.
2. Fit the netlist to measured impedance and drum response of the reference headphone, inspect the identifiability report, and export the fitted design.
3. Export the ANC plant with its uncertainty set and delay budget for a controller designer.
4. Derive production limit curves from the tolerance envelope and export them for end-of-line test.
5. Compare a design against two targets on two fixtures and see why the numbers differ.

19. Implementation plan and phasing

The first release is a linear, frequency-domain workbench at levels L0 and L1 with standards-based ear loads, honest validity bands, model-derived audition and a validation suite that runs in continuous integration. Nothing nonlinear, no in-browser finite elements and no controller design ship in it. That release already exceeds every existing headphone tool in the review of Section 20 on ear-load fidelity and validation, and it is small enough to finish.

Personas and acceptance. The acoustic engineer tunes a vent to a bass target within excursion and isolation constraints and checks it across fits. The DSP and ANC engineer exports the plant with its uncertainty set and delay budget. The quality engineer exports limit curves. The DIY designer and student use presets, the explain panel and audition. Each phase below states what those users can do and how it is tested.

Phase 1, first release

1. Rust to WebAssembly double-precision engine: netlist to stamps, equilibrated dense complex LU with condition monitoring and DC handling, adaptive log grid, Web Worker execution with a message-passing fallback, text netlist as the canonical file plus the schematic view.
2. Element library of Section 6 with the thermoviscous ladders, cavities, vents with per-end corrections and mesh, meshes and perforates from datasheet values, porous fill and pad paths with compression, the leak sub-network with standardised presets, radiation loads, tabulated user elements.
3. Driver at levels D1 and D2 with the datasheet-anchored reference driver and generic class presets; drive level per IEC 60268-7.

4. Ear loads: the 60318-4 transfer-matrix chain fitted to the standard's table with the damped variant, the Type 4.3 canal fitted to the standard's public geometry and transfer-impedance target, flat-plate and free-field; drum output with human-valid and coupler-only labelling; field-related overlays tagged by fixture.
5. Outputs: drum pressure, impedance magnitude and phase, excursion against the limit, group delay and excess phase, passive insertion loss from the same matrix, sensitivity in both conventions, finite-difference tornado charts reusing the factorisation, Monte Carlo bands over leak, front volume, clamp force, mesh and driver tolerances, and the advisory level flags.
6. Typed targets with the CC-BY 5128 curves bundled, parametric approximations flagged, the metric set, and the preference score greyed unless the fixture matches.
7. Fitting on impedance plus response with the identifiability report and confidence intervals; importers for REW, FRD, ZMA, CSV and WAV.
8. Audition: re-solved on the audio grid, excess-phase policy, partitioned convolution in an AudioWorklet with the native fallback, loudness matching, relative A/B, true-peak limiter, labelled not calibrated SPL.
9. Validation in continuous integration: every analytical oracle of Section 17, the 60318-4 table, the Type 4.3 target, and the open reference headphone once measured.
10. Project schema with migrations, report export as HTML and PDF, CSV, FRD, WAV and SOFA exports, and the licence register.

Phase 1 acceptance. The 60318-4 network passes the standard's table at all 21 frequencies. The reference driver's datasheet impedance is reproduced within 3 percent. The reference headphone is within 2 dB to 1 kHz and 4 dB from 1 to 4 kHz after fitting leak and front volume. The audition chain is within 0.1 dB of the analytic response. A 64-node, 512-point solve completes in under 50 ms, a target until the benchmark milestone confirms it.

Phase 2. The linear ANC panel with plant export, delay budget, Bode, Nyquist and sensitivity plots and IEC 60268-24 and ETSI TS 103 640 reporting; hardening of the L1 modal cup model against a measured cup; the companion solver in its first form, meaning automatic meshing and boundary generation from the parametric geometry, 2-D axisymmetric and coarse 3-D Helmholtz finite elements with boundary-layer losses, the verification gates, and deployment as a single container on the local network with browser discovery; production limit export and the quality-control view; DSP correctability metrics; multi-driver crossovers; adjoint sensitivities; the manufacturing tolerance framework with left-right tracking.

Phase 3. Nonlinear time domain with distortion trends and the thermal model; the in-browser 2-D axisymmetric finite elements with reduced-order sweeps for small problems; the companion solver's second form, boundary-element exterior radiation and pinna scattering, fine three-dimensional sweeps, and offloading of large Monte Carlo and optimisation batches; planar and electrostatic plug-ins validated against measured units; hear-through and occlusion; headband mechanics; the scripting API and command-line runner; feedforward coherence and causality analysis; the separable listening-test and preference-elicitation module.

Product roadmap and exclusions. Circumaural first. Supra-aural follows as the first variant because it reuses the leak sub-network and the 60318-1 coupling. Open wearable stereo follows with priority over in-ear: it drops the enclosed chamber, so the ear load becomes the open canal with its radiation impedance on a head, bass comes from standoff and dipole balance, bleed is a design output rather than a nuisance, and natural pinna incidence is the whole point, which makes it the first user of the companion solver's exterior boundary elements and the pinna-panel objective. In-ear follows last with the canal residual volume, nozzle and tip models. Out of scope throughout: in-browser three-dimensional boundary or finite elements, individual-ear prediction claimed as such, calibrated absolute-SPL audition, and adaptive noise-control simulation beyond an optional worker job.

Technology. In the browser: Rust compiled to WebAssembly with SIMD for numerics, a benchmarked choice between a linear-algebra crate and a hand-written small complex LU because the crate's WebAssembly behaviour is unconfirmed, pure-Rust FFTs, and wasmbindgen; TypeScript for the netlist model, interface and Web Audio; Web Workers everywhere with shared memory only when the page is cross-origin isolated; Canvas 2D for plots and WebGL2 for Monte Carlo fans; origin private file system and IndexedDB for storage. The companion solver: a Python service built on gmsh, scikit-fem or FEniCSx, PETSc with MUMPS and Bempp-cl, packaged as one container, reached over HTTP and WebSocket on the same machine or the local network, sharing the JSON contracts and the validation oracles with the browser engine so both are tested against the same cases. Everything is free and open source; no commercial solver is a dependency at any level. Rust tests run natively and under wasm32, Python tests against the same oracles, property-based tests in all three languages, golden files compared with tolerances.

Documentation. Every element carries a physics card with its equations, symbols and units, validity limits, references and the tests that cover it. A manual chapter explains validity shading and Monte Carlo bands, because those are the two things a first-time user misreads.

Risks and how they are handled

Risk	Handling
Performance budgets are unbenchmarked	Stated as targets; a benchmark milestone precedes any interactivity promise, especially for L2 and the companion solver
No public material data for pads, fabrics and fleece	A measurement campaign to ISO 9053 and ISO 10534-2 seeds the library; estimated entries are badged until replaced
Reference headphone needs fixture time on 60318-4 and Type 4.3 rigs	Partner laboratory; publish geometry and data under CC-BY
Standards tables are copyrighted	Fit models and cite; never embed tables; several clause numbers differ between editions and are quoted only after the text is read
Planar and electrostatic models lack peer-reviewed references	Experimental status until validated against a measured unit
Browser variation: shared memory and render quantum hints are not universal, Safari evicts storage	Feature detection with fallbacks; export warnings
Scope creep toward listening tests and adaptive control	Specified as separable modules with their own acceptance criteria
Lumped phase is wrong exactly where feedback ANC crosses over	An L1 or measured plant is required for stability conclusions, enforced in the panel

20. References and standards

Sources were located and, where possible, opened during preparation of this specification in September 2026. Items marked unverified were not retrieved in full or had details that could not be confirmed; they must be checked before their numbers or clause references enter code. Standards tables are copyrighted and are fitted or cited, never embedded.

Standards

- IEC 60318-4:2010, Occluded-ear simulator for insert earphones (formerly IEC 60711:1981). IEC 60318-1:2009, Ear simulator for supra-aural and circumaural earphones. IEC TS 60318-7:2017, Head and torso simulator (a later edition is unverified). IEC 60318-8:2022, High-frequency coupler for hearing aids and insert earphones (values not obtained).
- IEC 60268-7:2025 Ed. 4, Headphones and earphones, published 2025-06-27, superseding IEC 60268-7:2010 with Amendment 1:2020; clause numbers quoted only after reading. IEC 60268-24:2023, Active acoustic noise cancelling characteristics. IEC 62458, Large-signal parameters (content not retrieved). IEC 61094-4, Working standard microphones. IEC 61260-1:2014, Fractional-octave filters. IEC 61672-1, Sound level meters. IEC 61938 (1996), source-impedance recommendation, attribution of the 120 Ω figure unverified.
- ITU-T P.57 (06/2021), Artificial ears. ITU-T P.58 (06/2021, with a 03/2023 edition), Head and torso simulator for telephonometry. ITU-T P.380, Headsets. ITU-T H.870 (03/2022), Safe listening.
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21. Appendices

A. Glossary and units

Term	Meaning	Unit
DRP	Drum reference point: pressure at the drum-simulator microphone plane	Pa, dB SPL
EEP	Ear canal entrance point, open or blocked	Pa

Term	Meaning	Unit
ERP	Ear reference point, used for handsets; not a headphone output	Pa
PDR	Pressure division ratio, open over blocked entrance pressure	dimensionless
FEC	Free-air equivalent coupling: a headphone whose source impedance matches the open ear's radiation impedance	qualitative
kL	Wavenumber times characteristic length; lumped validity measure	dimensionless
Specific flow resistance	Pressure drop per unit face velocity of a mesh or fabric	MKS rayl, Pa·s/m
Flow resistivity	Specific flow resistance per unit thickness of a porous material	Pa·s/m ²
Acoustic impedance	Pressure over volume velocity	Pa·s/m ³
Acoustic inertance	Effective moving air mass of an aperture	kg/m ⁴
Acoustic compliance	Volume per unit pressure	m ³ /Pa, equal to m ⁵ /N
Bl	Force factor of a moving-coil motor	T·m, equal to N/A
Sd	Effective diaphragm area	m ²
Characteristic voltage	Voltage producing 94 dB SPL at 500 Hz at DRP through the specified source impedance	V
LKFS, LU	Loudness per ITU-R BS.1770 and loudness units relative to a reference	dB-like

B. Netlist schema example

```
{
  "schema": "acoustilab-netlist/2.0",
  "units": "SI-with-suffixes",
  "air": { "T_C": 23, "P_Pa": 101325, "RH": 0.5 },
  "tier": 1,
```

```

    "sweep": { "f_min_Hz": 10, "f_max_Hz": 40000, "points_per_octave": 48,
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    "nodes": [
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      { "id": "m_dia", "domain": "mechanical" },
      { "id": "a_front", "domain": "acoustic", "cavity": { "geometry": "box",
    "Lx_mm": 60, "Ly_mm": 45, "Lz_mm": 20, "liner": "fleece_A" } },
      { "id": "a_rear", "domain": "acoustic", "cavity": { "geometry":
    "cylinder", "a_mm": 25, "d_mm": 25 } },
      { "id": "a_canal", "domain": "acoustic" }
    ],
    "elements": [
      { "id": "amp", "type": "vsource", "nodes": ["e_in", "e_gnd"], "drive":
    "characteristic_voltage", "Zout_ohm": 0 },
      { "id": "coil", "type": "lossy_inductor", "nodes": ["e_in", "e_coil"],
    "Re_ohm": 32, "K": 0.02, "n": 0.7 },
      { "id": "motor", "type": "gyrator", "nodes": ["e_coil", "e_gnd", "m_dia",
    "m_gnd"], "B1_Tm": 1.0, "B1_x_poly": [1.0, -0.5, -40] },
      { "id": "susp", "type": "mech_rlc", "node": "m_dia", "Mms_g": 0.2,
    "Cms_mm_per_N": 1.0, "Rms_Ns_per_m": 0.05, "creep_lambda": 0.03 },
      { "id": "piston", "type": "transformer", "nodes": ["m_dia", "a_front",
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      "modes": [ { "f_Hz": 4500, "zeta": 0.03, "shape": "dome_1" } ] },
      { "id": "front_mesh", "type": "screen", "nodes": ["a_front", "a_canal"],
    "material": "mesh_070", "area_cm2": 12 },
      { "id": "pad_leak", "type": "slit_lrf", "nodes": ["a_front", "a_amb"],
    "h_mm": 0.3, "w_mm": 40, "l_mm": 8,
      "distribution": { "h_mm": { "lognormal": { "median": 0.3, "sigma": 0.6
    } } } },
      { "id": "baffle_vent", "type": "tube_lrf", "nodes": ["a_front",
    "a_rear"], "d_mm": 2, "L_mm": 1.5, "count": 4, "mesh": "mesh_260" },
      { "id": "rear_vent", "type": "tube_lrf", "nodes": ["a_rear", "a_amb"],
    "d_mm": 3, "L_mm": 2, "count": 6, "radiation": "flanged" },
      { "id": "ear", "type": "macro:iec60318_4", "nodes": ["a_canal"],
    "variant": "damped_hf", "mic_equiv_volume_mm3": 15 }
    ],
    "probes": [
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      { "id": "zin", "quantity": "impedance", "element": "amp" },
      { "id": "x", "quantity": "displacement", "node": "m_dia" },
      { "id": "U_leak", "quantity": "volume_velocity", "element": "pad_leak" }
    ]
  }

```

Suffixes on keys carry units so a misread file fails loudly. Every element type has a JSON schema with ranges, and the assembler rejects values outside them.

C. Derivations behind the validation checks

Air at 20 °C: density 1.204 kg/m³, speed of sound 343 m/s, so ρc^2 is 1.4165e5 Pa; viscosity 1.81e-5 Pa·s; ratio of specific heats 1.4; Prandtl number 0.71.

C1. Pressure chamber. Compressing a sealed volume V by a piston of area S_d moving x gives

$$p = \gamma P_0 \frac{S_d x}{V} = \rho c^2 \frac{S_d x}{V}$$

Below the coupled resonance the diaphragm is stiffness-controlled, x equals $BI \cdot U / (Re \cdot K)$ with K the total stiffness, so x and therefore p are independent of frequency down to DC. For 10 cm² and 30 cm³, one micron gives 4.72 Pa, or 107.5 dB SPL.

C2. Coupled resonance and high-frequency fall.

$$f_c = \frac{1}{2\pi} \sqrt{\frac{1/C_{ms} + S_d^2/C_{af}}{M_{ms}}}, \quad f_c^2 = f_s^2 + \frac{S_d^2}{4\pi^2 M_{ms} C_{af}}$$

The second form extracts the front compliance from a measured resonance shift. Above the resonance x falls as 1 over ω squared and so does p , giving 12 dB per octave. With 10 cm², 30 cm³ and 1 mm/N, a 0.1 g diaphragm resonates at 1204 Hz and a 0.3 g one at 695 Hz; the cavity stiffness of 4722 N/m dominates the suspension, so suspension compliance is a weak parameter in a sealed design and front volume a strong one.

C3. Leak slopes. A volume-velocity source into a compliance C shunted by leak impedance Z gives, relative to the sealed case,

$$H(\omega) = \frac{Z_L}{Z_L + 1/(j\omega C)}$$

A resistive leak yields a first-order high-pass with corner $1/(2\pi RC)$ and 6 dB per octave. An inertive leak yields a second-order high-pass with corner $1/(2\pi v(MC))$, 12 dB per octave and a Helmholtz peak. A real slit has both. Narrow-slit limits for gap h , width w and depth l :

$$R = \frac{12\mu l}{w h^3}, \quad M = \frac{6}{5} \frac{\rho l}{w h}$$

The exact laminar solution uses an effective density

$$\rho_{\text{eff}} = \frac{\rho}{1 - \tanh(kh/2)/(kh/2)}, \quad k = \sqrt{j\omega\rho/\mu}, \quad Z_L = \frac{j\omega\rho_{\text{eff}}}{1} w h$$

The wrong branch of the square root gives negative resistance, which the passivity test catches. The viscous boundary layer $\sqrt{2\mu/\rho\omega}$ is 0.49 mm at 20 Hz, 0.22 mm at 100 Hz and 0.069 mm at 1 kHz, so a 1 mm eyeglass gap is inertive across the audio band and resonates with 30 cm³ near 550 Hz with a Q well below the Poiseuille estimate, while a 0.2 mm gap is resistive with a corner at 83 Hz and no resonance.

C4. Helmholtz vent.

$$f_H = \frac{c}{2\pi} \sqrt{\frac{S}{V L_{\text{eff}}}}, \quad L_{\text{eff}} = L + \Delta_1 + \Delta_2$$

Each flanged end adds $8a/(3\pi)$, about 0.85a; an unflanged end adds 0.61a after Levine and Schwinger. A 3 mm hole in a 2 mm wall on 100 cm³ tunes to 215 Hz with both ends flanged and 325 Hz with no correction. Neck losses use the circular-tube thermoviscous solution, and radiation resistance is negligible at this scale.

C5. Electrical impedance.

$$Z_{\text{in}} = R_e + Z_L(\omega) + \frac{(Bl)^2}{Z_m}, \quad Z_m = R_{\text{ms}} + j\omega M_{\text{ms}} + \frac{1}{j\omega C_{\text{ms}}} + S_d^2 \frac{1}{Z_{\text{a,front}} + Z_{\text{a,rear}}}$$

In free air the peak height is R_e plus Bl^2/R_{ms} . A lossless cavity moves the peak to the coupled resonance without changing its height; acoustic loss lowers and broadens it, which is the practical diagnostic for seal and fill.

C6. Fill thermodynamics. Isothermal compliance is γ times adiabatic, so the limit is plus 40 percent. With fill fraction ϕ and thermal relaxation time τ ,

$$C_A(\omega) = C_{\text{ad}} \left[1 + \frac{(\gamma-1)\phi}{j\omega\tau} \right]$$

The imaginary part is a thermal loss. A fixed plus 12 percent corresponds to ϕ of 0.30 at low frequency only.

The unfilled cavity carries its own wall correction from the thermal boundary layer: compliance rises by ϵ and gains a loss tangent ϵ , with ϵ equal to $(\gamma - 1)$ times the thermal layer thickness times wall area over twice the volume, the complex form with the

factor $(1 - j)$ over 2. For a 30 cm³ cavity with 60 cm² of wall that is about 2.3 percent at 20 Hz, so an empty cavity is adiabatic to good approximation but never lossless, and its Q of 1 over ϵ lies between 30 and 200.

C7. Sensitivity scaling. At constant coil mass, resistance scales with turns squared and Bl with turns, so Bl is proportional to $\sqrt{Re}$. Pressure is proportional to $Bl \cdot V / Re$, hence to $V / \sqrt{Re}$ at constant voltage: 32 to 300 Ω is minus 9.72 dB. At constant power the level is unchanged. Conversion: dB per volt equals dB per milliwatt plus $10 \log_{10}(1000/Re)$.

C8. Lumped validity. A closed duct driven at one end has far-wall pressure $1/\cos(kL)$ times the driven pressure, so the lumped error is about $(kL)^2/3$: under 3 percent below kL of 0.3, 3 dB at kL of 1. Rigid-box modes:

$$f_{lmn} = \frac{c}{2} \sqrt{\left(\frac{l}{L_x}\right)^2 + \left(\frac{m}{L_y}\right)^2 + \left(\frac{n}{L_z}\right)^2}$$

Rigid-cylinder modes use Bessel derivative zeros 1.8412 for the first azimuthal and 3.8317 for the first radial mode:

$$f = \frac{c}{2\pi} \sqrt{\left(\frac{\alpha_{mn}}{a}\right)^2 + \left(\frac{q\pi}{d}\right)^2}$$

C9. Bleed estimate. The rear-side volume velocity radiates as a monopole with pressure $p_f|U|/(2r)$ at distance r , reduced below k_d of 1 by the dipole factor k_d for front and rear openings separated by d . For a sealed 30 cm³ front at 94 dB, the monopole bound at 1 m is 32 dB at 1 kHz and 60 dB at 5 kHz; real open-backs are leakier and land 40 to 60 dB below ear level in the midrange, so the output is a band, not a line.

D. Thermoviscous two-port

For a uniform duct of hydraulic radius r and length l , with $F(z)$ equal to $\tanh(z)/z$ for slits and $2J_1(z)/(zJ_0(z))$ for circles:

$$\rho_{eff} = \frac{\rho}{1 - F(k_v r)}, \quad \frac{1}{K_{eff}} = \frac{1 + (\gamma - 1)F(k_t r)}{\gamma P_0}, \quad k_v = \sqrt{j\omega\rho/\mu}, \quad k_t = k_v \sqrt{\Pr}$$

$$\Gamma = j\omega \sqrt{\rho_{eff}/K_{eff}}, \quad Z_c = \frac{\sqrt{\rho_{eff}K_{eff}}}{S}, \quad T = \begin{bmatrix} \cosh \Gamma l & \\ & Z_c \sinh \Gamma l \end{bmatrix} \quad \begin{bmatrix} 1 & \\ & Z_c \end{bmatrix}$$

The low-frequency limits recover the Poiseuille resistances and the $6/5$ and $4/3$ mass factors; the high-frequency limit gives resistance growing with the square root of frequency and the geometric mass. A reciprocal two-port is stamped as an admittance block

$$Y = \frac{1}{B} \begin{bmatrix} D & -1 \\ -1 & A \end{bmatrix}$$

which is why lossless tubes are forbidden: B vanishes at every half-wave frequency.